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Analysis of the Generative Learning of Diffusion Models through Statistical Mechanics and Complex Systems Theory

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Abstract

This work addresses the problem of generation, which can be interpreted as learning the true underlying distribution of data starting from its empirical realization, known as the *dataset*. In this context, we focus on the analysis of **Diffusion Models**, introduced in the literature over the last decade, which exploit a stochastic dynamic to controllably inject noise into data and subsequently learn the inverse generative process. We will show how such models make it possible to obtain an estimate of the underlying distribution, enabling the extraction of new configurations consistent with the statistical structure of the observed data. The analysis focuses in particular on understanding the generative mechanism through the language of statistical mechanics and complex systems theory, with the aim of characterizing the regimes in which effective learning of the distribution occurs. In general, in fact, the goal is not simply to reproduce the observed samples, but rather to generate new and non-trivial configurations consistent with the original distribution. This aspect is studied through phase transition theory, analyzing the behavior of the system as the sample size and the dimensionality of the data space vary. Finally, the discussion extends the theoretical results obtained under weaker hypotheses, accompanying them with computational validation, in order to confirm the correctness and robustness of the analytically derived mechanisms.

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Chapter 1

Introduction

In this work we intend to address a central issue of our times, namely the generative question, which in its modern and computational form takes shape in *Artificial Intelligence*. We can view this as the need to produce new forms of originality, and thus new questions that in a contemporary key often demand prompt answers (**LLM** *Large Language Models*). Its recent appeal is probably dictated by the context in which we live, where the amount of available information is so vast that it requires not merely access or consumption, but above all continuous reprocessing. It is precisely within this perspective that our discussion is situated, one which will not aim at achieving total completeness in the information presented, but rather hopes that it may serve as a stimulus for posing new questions, or rather *prompts*, that are more original and, above all, continually renewed, since: "*The important thing is not to stop questioning. Curiosity has its own reason for existing.*" (**Einstein**).

The first issue we intend to address concerns a more precise definition of the "generative problem." Informally, we can interpret it as the attempt to obtain new possible configurations of reality. More formally, generating means being able to describe the underlying structure of the data, that is, its probability distribution, understood as a (suitably normalized) function. Starting from it, it then becomes possible to sample, that is, to extract new data or configurations consistent with such a structure. In the modern literature there are many models that pursue this goal, starting from the most disparate premises. For example, architectures such as the *Multi-layer Perceptron* arise with the purpose of approximating functions, or *Transformers*, which instead arise to optimally model sequences. What we will analyze in particular is a model introduced over the last decade, known as **Diffusion Models**.

We will trace the historical path that led to what today is their modern formulation, which thus actually allows us to obtain a possible answer to what was our starting problem. However, we will not analyze their very latest versions (known, for instance, as *Flow Matching*) that attempt to compete with the current state of the art in generation; rather, we will analyze the fundamental mechanisms underlying their generative process. Indeed, we start from the premise that "*What I cannot create, I do not understand,*" as Professor **Feynman** said, implying that to truly fully understand something, we must be able to recreate it.

Moreover, to make the reading accessible to different types of readers, we have

preliminarily introduced the notions of stochastic calculus and diffusion processes that are essential for the discussion, hoping that they will not prove redundant for those already familiar with the topic. We will then focus on the analysis of the models themselves, before arriving at the core of the narrative: the study of learning regimes through phase transition theory and the related computational verifications.

Having clarified the themes and objectives of the present work, we now introduce the analytical path of the thesis, in the hope that it will not be limited to a technical description of generative models, but will also help to highlight their conceptual implications, hoping that they may be a source and stimulus for future questions.

1.1 Review of Stochastic Calculus

In light of the above, it is essential to define what a diffusion process is. If we wanted to model the evolution of a signal sampled from a specific probability distribution, it is necessary to formalize the dynamics that allow this. Such an evolution is intrinsically noisy, since it is driven by a continuous stochastic process (such as Brownian motion) that progressively corrupts the structure of the data; it is therefore naturally modeled by means of a Stochastic Differential Equation (SDE). In particular, the forward dynamics that we seek in the generative setting must be such as to transform, in finite time, the complex empirical distribution of the data into a simple and tractable prior distribution (typically an isotropic standard Gaussian), so as to be able to simulate the time-reversed SDE to generate new samples starting from pure noise. To fully understand this mechanism and analyze the subsequent considerations on the effectiveness of learning and its regimes of validity, we introduce below its mathematical formulation.

Definition 1.1.1 (Probability space). Let us begin by recalling the definition of a probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$ where $\mathcal{F}$ is a σ -algebra closed under complementation and countable union, and containing the certain event (and hence also the impossible one), which we assume to be:

1. **Complete**, meaning that for every null set and every one of its subsets, measurability still holds. In particular:

$$N \in \mathcal{F}, P(N) = 0, A \subseteq N \Rightarrow A \in \mathcal{F} \text{ and } P(A) = 0$$

2. **With a right-continuous filtration.** To formalize this property, we first define the σ -algebra representing the right limit of information at time t :

$$\mathcal{F}_{t+} := \bigcap_{s>t} \mathcal{F}_s.$$

Requiring the filtration to be right-continuous means imposing the equality $\mathcal{F}_t = \mathcal{F}_{t+}$. In practical terms, this ensures that the flow of information evolves smoothly, without informational "jumps" toward the future: the observer gains no surprising information by waiting an infinitesimal amount of time after t . Hence, everything that is knowable an instant immediately after t is already considered known exactly at time t .

Definition 1.1.2 (Brownian Motion). Let $W = (W_t)_{t \in [0, T]}$ be a standard m -dimensional Brownian motion adapted to the filtration $\{\mathcal{F}_t\}$, that is, a process taking values in $\mathbb{R}^m$ such that:

1. $P(W_0 = 0) = 1$;
2. the paths $t \mapsto W_t$ are a.s. continuous;
3. for every $0 \leq s < t \leq T$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$;
4. $W_t - W_s \sim \mathcal{N}(0, (t - s)I_m)$.

It follows that W is a zero-mean Gaussian process, with covariance $\mathbb{E}[W_t W_s^\top] = (t \wedge s)I_m$ and, in particular, $\mathbb{E}[\|W_t\|^2] = mt$, $\mathbb{E}[W^i W^j] = \delta_{ij}t$. From the independence and stationarity of the increments (points 3-4) it also follows that W enjoys the **Markov property**: the law of W_t conditioned on the entire history $W_{t_1}, \dots, W_{t_n}$ depends only on the most recent state W_{t_n} .

Definition 1.1.3 (Stochastic Differential Equation). Let the following functions be given:

- $f : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$, known as the **drift** of the equation, i.e. the vector field indicating the instantaneous mean velocity of the process, and thus the deterministic force guiding the trajectory
- $g : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^{d \times m}$ known as the **diffusive** term of the equation, i.e. a tensor field that can be seen as a linear transformation capturing the correlations of the fluctuations among the different components of the process.

It is assumed that f and g are measurable with respect to the Borel σ -algebra on $\mathbb{R}^d \times [0, T]$, i.e. a measurable set.

Consider now the following problem:

$$\begin{cases} dY_t = f(Y_t, t) dt + g(Y_t, t) dW_t, \\ Y_0 = y_0 \in \mathbb{R}^d, \end{cases} \quad (1.1)$$

The differential notation just presented, however intuitive and widely used, is in fact a formal abbreviation. Although Brownian motion has continuous paths, the intrinsically chaotic nature of its increments makes such paths nowhere differentiable with respect to time in the classical sense. The system acquires its full mathematical and operational meaning only when interpreted in its equivalent integral form:

$$Y_t = y_0 + \int_0^t f(Y_s, s) ds + \int_0^t g(Y_s, s) dW_s. \quad (1.2)$$

In this formulation, the first integration term is the standard Lebesgue integral, while the second is a stochastic integral, called the Itô integral, which models the accumulation of the random perturbation (*diffusion*), weighted instant by instant by the function g .

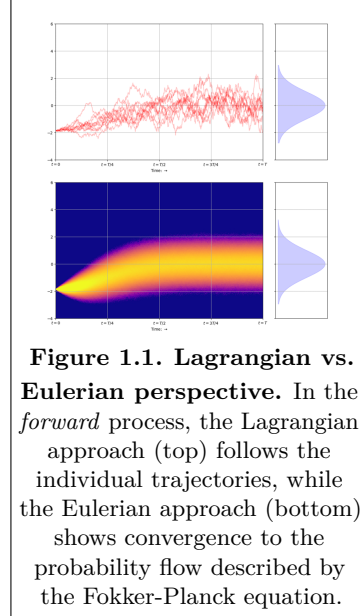
In general, for the integral form of the equation to admit a well-defined solution, regularity conditions must be imposed on the coefficients. A rigorous proof is

beyond the scope of the present work; we limit ourselves to stating the hypotheses of **Itô's Existence and Uniqueness Theorem** for the **strong solution**¹. Global Lipschitz continuity of f and g in x , uniformly in t ; 2. at most sub-linear growth of $|f|^2 + |g|^2$ in x ; 3. and initial data X_0 independent of the filtration generated by the Brownian motion, with finite second moment, $\mathbb{E}[|X_0|^2] < \infty$. Under these hypotheses, the SDE admits a unique strong solution.

Definition 1.1.4 (The Fokker-Planck Equation). Up to this point, the diffusion process has been described adopting a purely Lagrangian perspective, that is, describing the motion of the data points, so to speak. However, the founding goal of a generative model does not lie in the manipulation of individual samples, but rather in the ability to model, learn, and ultimately sample from the entire probability density of the data, as mentioned in (1). To make this conceptual leap, it is necessary to move to a Eulerian, i.e. macroscopic, formulation. It can be shown, as done in (1), that the equivalent Eulerian form of our SDE is the Fokker-Planck Equation, defined in the N -dimensional case as:

$$\frac{\partial P(\mathbf{x}, t)}{\partial t} = - \sum_{i=1}^N \frac{\partial}{\partial x_i} [f_i(\mathbf{x}, t) P(\mathbf{x}, t)] + \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2}{\partial x_i \partial x_j} [g_{ij}(\mathbf{x}, t) P(\mathbf{x}, t)], \quad (1.3)$$

However, in practice estimating this equation is unapproachable. If one were to try to discretize the spatial domain, dividing each axis into K intervals, the total number of grid nodes would explode to K^N . Even for very low-resolution images, such as the 28×28 samples of the MNIST dataset, this grid would require enormous computational effort. This phenomenon is known as the *Curse of Dimensionality*, for which a rigorous geometric justification is provided in Section 1, and it will be central to all our analyses.



1.2 Diffusion Process

The notions introduced in Section (1.1), the SDE, Brownian motion, and the Fokker-Planck equation, in fact describe the same object from two complementary points of view. We call a **diffusion process** a Markov process with continuous paths Y_t whose evolution, over an infinitesimal time interval dt , is completely determined by

¹By *strong* solution we mean the one obtained for a realization of the noise W_t fixed a priori, the only case we will refer to in this work.

only two quantities: a **drift** $f(Y_t, t)$, which describes the mean displacement of the process, and a **noise intensity** $g(Y_t, t)$, which determines its dispersion around the mean. Formally:

$$\mathbb{E}[dY_t | Y_t] = f(Y_t, t) dt, \quad \text{Var}[dY_t | Y_t] = g(Y_t, t)^2 dt,$$

that is, exactly the two terms appearing in the SDE (1.1.3). In other words, assigning the pair (f, g) in the differential notation $dY_t = f(Y_t, t) dt + g(Y_t, t) dW_t$ is equivalent, under the regularity conditions already discussed, to uniquely defining a diffusion process: the SDE provides its microscopic description (trajectory by trajectory), while the associated Fokker-Planck equation provides its macroscopic description (evolution of the density $p_t(x)$).

This equivalence is what we will use systematically in (2): in order to construct a forward process useful for generative purposes, it is not necessary to explicitly solve the SDE, but it suffices to choose f and g so that:

- the process transforms, as $t \rightarrow T$, any initial data distribution p_0 into a known limiting distribution that is easy to sample from (which, in standard models, is typically a Gaussian p_T);
- the transition density $p_t(Y_t | Y_0)$ is analytically tractable (e.g. Gaussian) and computable in closed form for every t , so as to be able to sample Y_t directly without having to numerically simulate the entire trajectory.

A particularly simple case, which will be the starting point of (2.1.1), is obtained by choosing f linear in Y_t and g constant: this is precisely the case of the Ornstein-Uhlenbeck process, which we will treat in its *Variance-Preserving* form in Section (2.1.1).

Chapter 2

Diffusion Generative Models

Before critically analyzing generative models based on Diffusion Processes, it may be interesting to retrace, from a historical point of view, how one arrived at defining them in this way, so as to gain a clearer picture of their behavior. One of the most intuitive statistical approaches for learning a probability distribution is to maximize the likelihood (Maximum Likelihood Estimation), i.e. to compute:

$$\max_{\theta} \mathbb{E}_{P_0(Y_0)}[\log P_{\theta}(Y_0)]$$

If we think of $P_{\theta}(Y_0) = \frac{\tilde{P}_{\theta}(Y_0)}{Z_{\theta}}$, as for instance in the case of Energy-based models where $\tilde{P}_{\theta}(Y_0) = e^{-E_{\theta}(Y_0)}$, then the normalization constant turns out to be Z_{θ} where $Z_{\theta} = \int \tilde{P}_{\theta}(Y) dY$, which, being an integral in dY with Y typically living in high dimension, is computationally inaccessible. Indeed, estimating Z has always been one of the most fascinating and treacherous challenges of statistical mechanics. Many approaches have been attempted to get around this obstacle, most famously the one theorized by Giorgio Parisi (1979) (the study of spin glasses and the replica method for the shape of the minima), which earned him the Nobel Prize in 2021.

Equally ingenious was the idea theorized by Aapo Hyvärinen in 2005 (Hyvärinen (2005)), who attempted to get around this obstacle by estimating a function called the **Score Function**, defined as the logarithmic gradient of the data density $\nabla_{Y_0} \log P_0(Y_0)$, which leads to learning the distribution without having to compute Z . In particular, Hyvärinen estimates a model $s_{\theta}(Y_0)$ by minimizing the Fisher Divergence:

$$\mathbb{E}_{P_0(Y_0)} \left[\frac{1}{2} \|s_{\theta}(Y_0) - \nabla_{Y_0} \log P_0(Y_0)\|_2^2 \right] \quad (2.1)$$

However, expanding the norm, that is:

$$\frac{1}{2} \|s_{\theta}(Y_0) - \nabla_{Y_0} \log P_0(Y_0)\|^2 = \frac{1}{2} \|s_{\theta}(Y_0)\|^2 - s_{\theta}(Y_0)^T \nabla_{Y_0} \log P_0(Y_0) + \frac{1}{2} \|\nabla_{Y_0} \log P_0(Y_0)\|^2,$$

for the purposes of the optimization problem with respect to θ , we can disregard the term that is constant in this variable (the third one), obtaining:

$$\mathbb{E}_{P_0(Y_0)} \left[\text{Tr}(\nabla_{Y_0} s_{\theta}(Y_0)) + \frac{1}{2} \|s_{\theta}(Y_0)\|^2 \right]. \quad (2.2)$$

where the trace was obtained by explicitly substituting the derivative of the logarithm and applying $\int s_{\theta,i}(Y) \frac{\partial P_0(Y)}{\partial Y_i} dY_i = \underbrace{[s_{\theta,i}(Y)P_0(Y)]_{-\infty}^{+\infty}}_{=0} - \int \frac{\partial s_{\theta,i}(Y)}{\partial Y_i} P_0(Y) dY_i$. As we

can see from this framework, the minimization problem does not depend on the gradient of the distribution, which is defined only in the vicinity of the data. However, we note that precisely the trace of the Jacobian would require computing the derivative d times (where d would be $64 \times 64 \times 3$), and is therefore inaccessible.

Then, in 2011 Pascal Vincent Vincent (2011) circumvents this problem by showing that if we consider Y_t , i.e. a version of Y_0 perturbed with noise added according to a transition probability $P(Y_t|Y_0)$, then maximizing the original objective is mathematically equivalent to minimizing:

$$\mathbb{E}_{P_0(Y_0)} \mathbb{E}_{P(Y_t|Y_0)} \left[\|s_{\theta}(Y_t) - \nabla_{Y_t} \log P(Y_t|Y_0)\|_2^2 \right] \quad (2.3)$$

thus being the first to introduce the formal concept of *Denoising Score Matching* (DSM). (The steps of the equivalence will be analyzed in (2.2.1); informally, we can think of having rewritten $P_0(Y_0)$ by means of Bayes' theorem.)

However, an implementation theory for effectively perturbing the data over time for generative purposes did not yet exist. In 2015 Jascha Sohl-Dickstein Sohl-Dickstein et al. (2015) conceives of creating an actual discrete process based on Markov chains. The choice of a Markovian framework faithfully models natural diffusion processes (such as Brownian motion), in which the probabilistic evolution of the system at a given instant is rigorously conditioned only on its current state. Considering thus $Y_t = \sqrt{1-\beta_t}Y_{t-1} + \sqrt{\beta_t}z$, with $z \sim \mathcal{N}(0, \mathbf{I})$, a process is thus introduced for t ranging from 1 to T , in which the probability of transitioning from one state to another (the *Forward Process*) is defined as:

$$q(Y_t|Y_{t-1}) = \mathcal{N}(Y_t; \sqrt{1-\beta_t}Y_{t-1}, \beta_t\mathbf{I}) \quad (2.4)$$

which is then estimated in the inverse process (*Backward Process*) by means of:

$$P_{\theta}(Y_{t-1}|Y_t) = \mathcal{N}(Y_{t-1}; \mu_{\theta}(Y_t, t), \Sigma_{\theta}(Y_t, t)). \quad (2.5)$$

Applying in his work the definition of the Loss via the ELBO (Evidence Lower Bound), which in the meantime had become the standard for training latent models thanks to Kingma and Welling (2013), namely:

$$\mathcal{L} = \mathbb{E}_q \left[-\log P_{\theta}(Y_T) - \sum_{t=1}^T \log \frac{P_{\theta}(Y_{t-1}|Y_t)}{q(Y_t|Y_{t-1})} \right] \quad (2.6)$$

This formulation, introduced by Sohl-Dickstein et al. (2015), is called **Variance Preserving**, since if $\text{var}(Y_{t-1}) = 1$ then: $\text{Var}(Y_t) = (1-\beta_t)(1) + \beta_t(1) = 1 - \beta_t + \beta_t = 1$. However, the data does not occupy the whole space $\mathbb{R}^d$ but rather a low-dimensional manifold, and the "empty space" surrounding it caused the numerical explosion of the gradients in the *forward* process. To solve this topological problem, Yang Song and Stefano Ermon (NCSN, 2019) Song and Ermon (2019) take up Vincent's DSM but introduce a scaling factor: instead of using a single noise level, they implement a geometric sequence $\sigma_1 > \sigma_2 > \dots > \sigma_L$, creating in discrete time a dynamic of

the type: $Y_t = Y_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} z$ with $z \sim \mathcal{N}(0, \mathbf{I})$, where we observe that the term under the square root is always > 1 . This formulation is generally known as **Variance Exploding** since $\text{Var}(Y_t) = \text{Var}(Y_0) + \sigma_t^2 \approx 1 + \sigma_t^2$.

Despite these insights, the mathematics of Sohl-Dickstein et al. (2015) was still too unstable to train at large scale, while the score matching of Song and Ermon (2019), due to the variance explosion, proved difficult to apply. The decisive unification came in 2020 with Jonathan Ho and the Denoising Diffusion Probabilistic Models (DDPM) Ho et al. (2020). Ho shows that optimizing the variational problem (ELBO) of Sohl-Dickstein and the Denoising Score Matching of Vincent/Song are, under specific (Gaussian) parametrizations, the same problem, an aspect we will analyze carefully later, and which is proven in (4).

All these methods, however, depended on a discrete number of time steps. In 2021, Yang Song Song et al. (2021) shows that, by taking the number of steps to infinity, or $\Delta t \rightarrow 0$, the *forward* process becomes a Stochastic Differential Equation (SDE) defined in continuous time $t \in [0, T]$:

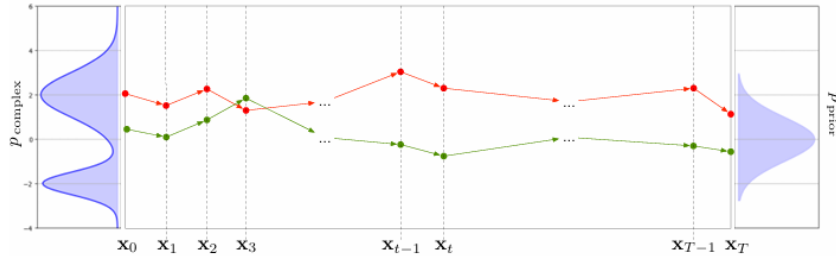
$$dY_t = f(Y_t, t)dt + g(Y_t, t)dW_t$$

and, applying Anderson’s theorem, he then derives the exact reverse SDE needed for generation:

$$dY_t = [f(Y_t, t) - g(Y_t, t)^2 \nabla_{Y_t} \log P_t(Y_t)]dt + g(Y_t, t)d\bar{W}_t.$$

In this equation, $\nabla_{Y_t} \log P_t(Y_t)$ is exactly the **Score Function** that the neural network approximates, and we will see that the **Variance-Preserving** formulation, by suitably defining the *drift* and noise terms, gives us an exact form with which to inject it (it will be a $\mathcal{N}(0, \mathbf{I})$). We will now go on to see the mathematical foundations necessary to rigorously understand these two processes, *Backward* and *Forward*, and how, through them, we manage to learn the distribution of the real data, and hence how the learning of the latter is influenced by various factors such as duration, data dimensionality, the chosen architecture, and so on.

2.1 Forward Process



In this section we will formalize the **Forward** process, that is, the process in which we theoretically progressively inject noise into the initial data distribution in order to artificially expand its spatial support, making the score function differentiable and well-defined at every point of the domain, but which nevertheless in practice we

should not use, since all we need is the form we obtain in the limit, which allows us to have an exact definition of the noise from which to start the *Backward* process, as mentioned earlier and as we will see in this chapter in greater detail. To make things somewhat clearer, consider a dataset $\{x_1, \dots, x_n\}$, that is, the set of data available to us. If we want to define its initial distribution $P_0(y_0)$, we obtain the form:

$$P_0(y_0) = \frac{1}{N} \sum_{i=1}^N \delta(y_0 - x_i), \quad (2.7)$$

that is, the normalized sum of Dirac deltas centered on the data points, which turns out to be a strongly non-differentiable and discontinuous function. However, if we recall what was introduced earlier, in order to solve a generative task, in the **Backward** process we must estimate the **Score Function**, i.e. $\nabla_{Y_t} \log P_t(Y_t)$. To obtain a sensible estimate of this operator we necessarily need regularity hypotheses on the form of $P_t(Y_t)$ (those assumed in (1.1)), which at the moment we therefore lack. A workable idea is to let the individual initial data points evolve according to a forward-in-time Stochastic Differential Equation (SDE), for $t \in [0, T]$, of the type:

$$\begin{cases} dY_t = a f(Y_t, t) dt + g(Y_t, t) dW_t, \\ Y_0 = y_0 \in \mathbb{R}^d, \end{cases} \quad (2.8)$$

where $f(Y_t, t)$ represents the *drift* coefficient (a deterministic contraction), $g(t)$ governs the amplitude of the diffusion, and W_t is a standard Brownian motion that progressively injects stochastic noise into the system.

This evolution constitutes the foundation of the **Forward Process**, since the continuous injection of Gaussian noise acts mathematically as a convolution on the initial distribution $P_0(Y_0)$, namely:

$$P_t(Y_t) = \int dY_0 P_t(Y_t|Y_0) P_0(Y_0) = \int \mathcal{K}_t(Y_t - Y_0) \left[\frac{1}{N} \sum_{i=1}^N \delta(Y_0 - x_i) \right] dY_0. \quad (2.9)$$

whose explicit steps can be found in (1). However, applying the linearity of the integral and the properties of the Dirac delta, we obtain that:

$$P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \mathcal{N}(Y_t; x_i, \sigma_t^2 \mathbf{I}) \quad (2.10)$$

that is, a sum of Gaussians known as a Gaussian Mixture Model (**GMM**). Note how this is a C^∞ function on all of $\mathbb{R}^D$, $\forall t > 0$, because it is a sum of C^∞ Gaussian functions. It is furthermore important to observe that this distribution is in general not Gaussian, but rather a multimodal distribution, i.e. with many peaks, which preserves the topology of our dataset.

2.1.1 The Variance-Preserving Ornstein-Uhlenbeck Process

In the original formulation of Diffusion Models Ho et al. (2020), a specific case of (2.8) is analyzed, namely the Ornstein-Uhlenbeck stochastic process, which, in general terms, takes the form:

$$\begin{cases} dY_t = q(m - Y_t)dt + \sigma dW_t \\ Y_0 = y_0 \end{cases} \quad (2.11)$$

where the linear drift term $q(m - Y_t)$ acts as a restoring force toward the mean value m , balancing the stochastic noise component driven by the Brownian motion dW_t . In the study of generative models, a constant parameter $\beta > 0$ is defined (commonly known as the *noise schedule*), and a specific parameter mapping is applied to obtain the **Variance-Preserving (VP)** formulation:

$$\begin{cases} q = \frac{1}{2}\beta \\ m = 0 \\ \sigma = \sqrt{\beta} \end{cases} \quad (2.12)$$

This formulation is commonly traced back to a well-known physical process, by fixing the noise injection rate to a specific constant: $\beta(t) = \beta = 2$. With this choice, the drift coefficient q becomes exactly unitary (since $\frac{1}{2} \cdot 2 = 1$) and $\sigma = \sqrt{2}$. The system thus takes the form of the Langevin equation for a particle immersed in a thermostat and confined in a **harmonic potential** $V(Y) = \frac{1}{2}Y^2$, where the elastic restoring force is given by the negative gradient of the potential, $-\nabla V(Y) = -Y$:

$$\begin{cases} dY_t = -Y_t dt + \sqrt{2} dW_t \\ Y_0 = y_0 \end{cases} \quad (2.13)$$

Proposition 2.1.1 (Explicit Solution). The unique strong-form solution of equation (2.13) can be rewritten in explicit integral form as:

$$Y_t = y_0 e^{-t} + \sqrt{2} \int_0^t e^{-(t-u)} dW_u \quad (2.14)$$

Proof. To simplify the computation of the differential, we rewrite (2.14) by factoring the t -dependent term out of the integral:

$$Y_t = y_0 e^{-t} + \sqrt{2} e^{-t} \int_0^t e^u dW_u \quad (2.15)$$

Computing the differential, we obtain:

$$dY_t = -y_0 e^{-t} dt - \sqrt{2} e^{-t} \left(\int_0^t e^u dW_u \right) dt + \sqrt{2} e^{-t} e^t dW_t \quad (2.16)$$

Collecting the dt terms:

$$dY_t = - \left[y_0 e^{-t} + \sqrt{2} e^{-t} \int_0^t e^u dW_u \right] dt + \sqrt{2} dW_t \quad (2.17)$$

Substituting the term in square brackets with the definition of Y_t given in (2.15), we recover the starting equation:

$$dY_t = -Y_t dt + \sqrt{2} dW_t \quad (2.18)$$

□

Proposition 2.1.2 (Conditional Expected Value). Considering the integral solution of the stochastic process Y_t , its expected value can be expressed as:

$$\mathbb{E}[Y_t | Y_0 = y_0] = \mathbb{E}[y_0 e^{-t}] + \mathbb{E}\left[\sqrt{2} \int_0^t e^{-(t-u)} dW_u\right] \quad (2.19)$$

However, $\mathbb{E}\left[\sqrt{2} \int_0^t e^{-(t-u)} dW_u\right] = 0$. Intuitively, since $\int_0^t f(u) dW_u \approx \sum f(t_i) \cdot (W_{t_{i+1}} - W_{t_i})$, and since the increments of Brownian motion are independent of the past and have zero mean, the expected value of the product coincides with $\mathbb{E}[f(t_i)] \cdot \mathbb{E}[W_{t_{i+1}} - W_{t_i}]$, which is zero for the reasons stated in (1.1.2).

We thus obtain the closed form for the mean of the transition:

$$\mathbb{E}[Y_t | Y_0 = y_0] = y_0 e^{-t} \quad (2.20)$$

Proposition 2.1.3 (Variance). If we want to consider the variance $v(Y_t|Y_0)$:

$$v(Y_t|Y_0) = \text{Var}\left(y_0 e^{-t} + \sqrt{2} \int_0^t e^{-(t-u)} dW_u\right) \quad (2.21)$$

Recalling the rule for the variance of an affine transformation, which is invariant under translations (i.e. $\text{Var}(X + b) = \text{Var}(X)$ if the term is deterministic given the initial state), we obtain:

$$v(Y_t|Y_0) = \text{Var}\left(\sqrt{2} \int_0^t e^{-(t-u)} dW_u\right) \quad (2.22)$$

Exploiting the second-degree homogeneity of the variance and Itô's isometry (the integral having zero mean), we can rewrite:

$$v(Y_t|Y_0) = \mathbb{E}\left[\left(\sqrt{2} \int_0^t e^{-(t-u)} dW_u\right)^2\right] = 2 \int_0^t e^{-2(t-u)} du \quad (2.23)$$

Solving the integral, we obtain:

$$v(Y_t|Y_0) = 2 \left[\frac{e^{-2(t-u)}}{-2} \right]_0^t \quad (2.24)$$

that is,

$$v(Y_t|Y_0) = 1 - e^{-2t} \quad (2.25)$$

Proposition 2.1.4 (Transition Probability and Marginals). In the Variance-Preserving case, we rewrite equation (2.10), where in this case the transition probability from the data point $x_i = y_0$ to one of its possible evolutions Y_t is given by:

$$P(Y_t|Y_0 = x_i) = \mathcal{N}(Y_t; x_i e^{-t}, (1 - e^{-2t})\mathbf{I}), \quad (2.26)$$

where the conditional probability acts as a time-dependent Gaussian transition kernel $\mathcal{K}_t(Y_t, x_i)$, which, applied to an Ornstein-Uhlenbeck process, in addition to

adding noise as a classical convolution $\mathcal{K}_t(Y_t - x_i)$ would do, also multiplies by a decay factor e^{-t} , yielding the form:

$$\mathcal{K}_t(Y_t, x_i) = \frac{1}{(2\pi(1 - e^{-2t}))^{\frac{D}{2}}} \exp\left(-\frac{\|Y_t - x_i e^{-t}\|^2}{2(1 - e^{-2t})}\right). \quad (2.27)$$

This mathematical formulation makes explicit the delicate thermodynamic balance of the process: the kernel progressively blurs the information by injecting noise while, at the same time, applying a deterministic drift that contracts the space toward the origin (the decay term $x_i e^{-t}$).

Starting from the conditional transition probability, we can derive the marginal distribution, as done in (2.10), of the VP process at time t , defined as $P_t(Y_t)$. This distribution is obtained by averaging the initial density $P_0(y_0)$ over all possible trajectories, namely:

$$P_t(Y_t) = \int dY_0 P_t(Y_0, Y_t) = \int dY_0 P_t(Y_t|Y_0)P_0(Y_0) \quad (2.28)$$

from which, again following the steps in (1), we obtain:

$$P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \mathcal{N}(Y_t; x_i e^{-t}, (1 - e^{-2t})\mathbf{I}) \quad (2.29)$$

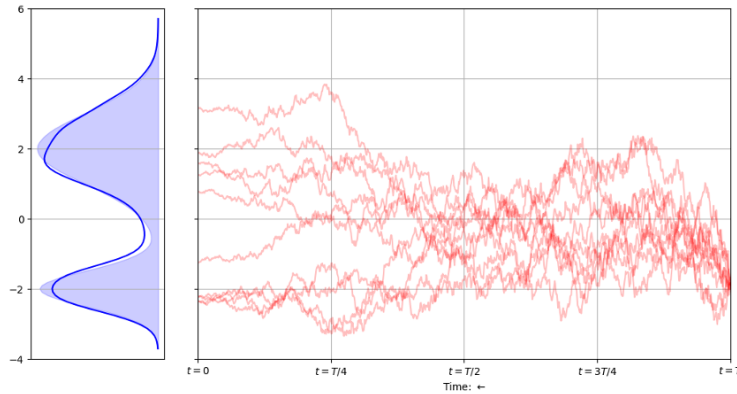
that is the Monte Carlo version of:

$$P_t(Y_t) = \int dY_0 P_0(Y_0) \frac{1}{\sqrt{(2\pi\Delta_t)^N}} \exp\left(-\frac{\|Y_t - Y_0 e^{-t}\|^2}{2\Delta_t}\right) \quad (2.30)$$

where $\Delta_t = 1 - e^{-2t}$.

In particular, if we consider kernels of the form (2.27), it can be shown that, as $t \rightarrow \infty$, the marginal distribution (2.30) converges to a $\mathcal{N}(0, \mathbf{I})$. The explicit steps of the proof can be found in (2).

2.2 Backward Process



Starting then from the elegant formulation of this process that we have arrived at in Song et al. (2021), obtained from the solution of the **Score Matching** problem, which was then combined with that of finding a suitable *denoiser*, we define, thanks to **Anderson’s Theorem** (1982), proven in (1), the following SDE:

$$dY_t = [-Y_t - 2\nabla_{Y_t} \log P_t(Y_t)]dt + \sqrt{2}d\bar{W}_t \quad (2.31)$$

with t in this case $\in [T, 0]$. The presence of the **Score** function, i.e. $\nabla_{Y_t} \log P_t(Y_t)$, is fundamental. Recalling that $P_t(Y_t|Y_0 = x_i) \sim \mathcal{N}(Y_t, x_i e^{-t}, (1 - e^{-2t})\mathbf{I})$, defined in (2.26), from which we obtained that $P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \mathcal{N}(Y_t, x_i e^{-t}, (1 - e^{-2t})\mathbf{I})$, which allowed us to rewrite the term as:

$$\nabla_{Y_t} \log P_t(Y_t) = \nabla_{Y_t} \log \left(\frac{1}{N} \sum_{i=1}^N \mathcal{N}(Y_t, x_i e^{-t}, (1 - e^{-2t})\mathbf{I}) \right) \quad (2.32)$$

However, if we look at this quantity, since we cannot in any way factorize the logarithm of the sum, it requires computing N sums for every time instant, which makes it inaccessible. If we wanted to try to provide an estimate of it via $s_\theta(Y_t)$, at time t , we would minimize the quantity:

$$\mathcal{L}_t(\theta) = \mathbb{E}_{P_t(Y_t)} [\|s_\theta(Y_t) - \nabla_{Y_t} \log P_t(Y_t)\|^2], \quad (2.33)$$

which, however, due to the analytical intractability of the **Score** term, is inaccessible; indeed, if we tried to expand the norm, we would obtain the trace of the Jacobian matrix, which, as stated in (2.1), is computationally inaccessible. However, we will observe in (2.2.1) that, by writing $P_t(Y_t) = \int dY_0 P_t(Y_t|Y_0)P_0(Y_0)$, we will manage to get around this computational bottleneck, obtaining an equivalence in the optimization problem as shown in Vincent (2011), yielding:

$$\mathbb{E} \left[\|\nabla_{Y_t} \log P(Y_t|Y_0) - s_\theta(Y_t)\|_2^2 \right], \quad (2.34)$$

In this way the problem of estimating the **Score Function** is transferred to the neural network, as we will see further on. With the introduction of autoencoders, this formulation of the cost function finds a concrete operational realization. Consider, for instance, the case in which the architecture does not reduce dimensionality, and let us further assume, as is typically the case, that the *denoiser* uses an L_2 -norm (**MSE**) loss, which minimizes the distance between the original data Y_0 and the function $f_\theta(Y_t)$, i.e. the output of the network applied to the perturbed data. The network, however, does not aim directly at estimating Y_0 , but rather at approximating the **score** function $s_\theta(Y_t) \approx \nabla_{Y_t} \log P_t(Y_t)$. To connect these two perspectives, we start from the result of Vincent (2011) and consider the loss in (2.34). Since $P_t(Y_t|Y_0) \sim \mathcal{N}(Y_t; x_i e^{-t}, (1 - e^{-2t})\mathbf{I})$, we can rewrite the conditional score as:

$$\nabla_{Y_t} \log P(Y_t|Y_0) = -\frac{Y_t - Y_0 \mu_t}{\sigma_t^2} = \frac{Y_0 \mu_t - Y_t}{\sigma_t^2} \quad (2.35)$$

from which we obtain:

$$\min_{\theta} \mathbb{E} \left[\left\| \frac{Y_0 \mu_t - Y_t}{\sigma_t^2} - s_\theta(Y_t) \right\|_2^2 \right]. \quad (2.36)$$

Multiplying or dividing by a constant at time t such as $\frac{\sigma_t^4}{\mu_t^2}$ does not change the optimization problem, and so we obtain:

$$\min_{\theta} \mathbb{E} \left[\left\| \frac{\sigma_t^2}{\mu_t} s_{\theta}(Y_t) - \frac{Y_0 \mu_t - Y_t}{\mu_t} \right\|_2^2 \right] \quad (2.37)$$

which we rewrite as:

$$\min_{\theta} \mathbb{E} \left[\left\| \frac{\sigma_t^2}{\mu_t} s_{\theta}(Y_t) - Y_0 + \frac{Y_t}{\mu_t} \right\|_2^2 \right] \quad (2.38)$$

which finally becomes:

$$\min_{\theta} \mathbb{E} \left[\left\| Y_0 - \left(\frac{Y_t + \sigma_t^2 s_{\theta}(Y_t)}{\mu_t} \right) \right\|_2^2 \right] \quad (2.39)$$

At this point we see that:

$$f_{\theta}(Y_t) = \frac{Y_t + \sigma_t^2 s_{\theta}(Y_t)}{\mu_t} = \frac{Y_t + (1 - e^{-2t}) s_{\theta}(Y_t)}{e^{-t}} \quad (2.40)$$

that is, the function the autoencoder attempts to approximate in terms of the **Score function** $s_{\theta}(Y_t)$. Since it is also known that for a problem with MSE loss the best predictor is precisely the conditional expectation, i.e. $\mathbb{E}[Y_0|Y_t]$, the optimal predictor of the Autoencoder therefore corresponds exactly to:

$$f_{\theta}^*(Y_t) = \mathbb{E}[Y_0|Y_t] \quad (2.41)$$

Substituting this identity into (2.39), and recalling that at the optimum the network $s_{\theta}(Y_t)$ converges to the true score of the marginal $\nabla_{Y_t} \log P_t(Y_t)$, we derive the well-known **Tweedie Formula** for an Ornstein-Uhlenbeck process:

$$\mathbb{E}[Y_0|Y_t] = \frac{Y_t + \sigma_t^2 \nabla_{Y_t} \log P_t(Y_t)}{\mu_t} = \frac{Y_t + (1 - e^{-2t}) \nabla_{Y_t} \log P_t(Y_t)}{e^{-t}}, \quad (2.42)$$

and, inverting it algebraically, allows us to write:

$$\nabla_{Y_t} \log P_t(Y_t) = \frac{\mu_t \mathbb{E}[Y_0|Y_t] - Y_t}{\sigma_t^2} \quad (2.43)$$

Thus obtaining a fundamental result, namely that training a network exactly for **Denoising Score Matching** is equivalent to training an optimal Autoencoder, with no dimensionality reduction in our case. This result is known as the **Tweedie** formula, and this equivalence was first introduced in Alain and Bengio (2014).

2.2.1 The Loss

For the treatment of the cost function we will use an approach based on the **Denoising Score Matching** framework. However, for those more familiar with the approach of Ho et al. (2020), i.e. the standard one used in *AutoEncoders*, its definition can be found in (3), and moreover, in Proposition 4, all the steps necessary

to obtain the equivalence with this framework can be found. Let us therefore consider the marginal loss at time t , which, for the reasons stated, takes the form:

$$\mathcal{L}_t(\theta) = \mathbb{E}_{P_t(Y_t)} \left[\|s_\theta(Y_t) - \nabla_{Y_t} \log P_t(Y_t)\|^2 \right], \quad (2.44)$$

thus obtaining $\mathcal{L}(\theta)$ as:

$$\mathcal{L}(\theta) = \int_0^T dt \mathbb{E}_{P_t(Y_t)} \left[\|s_\theta(Y_t) - \nabla_{Y_t} \log P_t(Y_t)\|^2 \right]. \quad (2.45)$$

We now want to expand (2.44), which we thus rewrite as:

$$= \underbrace{\mathbb{E}_{P_t(Y_t)} [\|s_\theta(Y_t)\|^2]}_1 - \underbrace{2\mathbb{E}_{P_t(Y_t)} [\langle s_\theta(Y_t), \nabla_{Y_t} \log P_t(Y_t) \rangle]}_2 + \underbrace{\mathbb{E}_{P_t(Y_t)} [\|\nabla_{Y_t} \log P_t(Y_t)\|^2]}_3, \quad (2.46)$$

observing that term (1) is easy for the purposes of the optimization problem, and seeks the network output with minimal L_2 norm. Term (3), moreover, is a constant in θ and so we neglect it. Term (2) can be rewritten explicitly in its integral form as:

$$\mathbb{E}_{P_t(Y_t)} [\langle s_\theta(Y_t), \nabla_{Y_t} \log P_t(Y_t) \rangle] = \int dY_t \langle s_\theta(Y_t), \nabla_{Y_t} \log P_t(Y_t) \rangle P_t(Y_t) \quad (2.47)$$

which, by computing the derivative of the logarithm, we can rewrite as:

$$\int dY_t P_t(Y_t) \langle s_\theta(Y_t), \frac{\nabla_{Y_t} P_t(Y_t)}{P_t(Y_t)} \rangle \quad (2.48)$$

and simplifying, using the homogeneity of the scalar product, we obtain:

$$\int dY_t \langle s_\theta(Y_t), \nabla_{Y_t} P_t(Y_t) \rangle, \quad (2.49)$$

moreover, recalling (2.30), by means of Bayes' Theorem, we can rewrite the term as:

$$\int \left\langle s_\theta(Y_t), \nabla_{Y_t} \left(\int P_t(Y_t|Y_0) P_t(Y_0) dY_0 \right) \right\rangle dY_t \quad (2.50)$$

and using the logarithm rule in reverse: $\nabla_{Y_t} P_t(Y_t|Y_0) = P_t(Y_t|Y_0) \nabla_{Y_t} \log P_t(Y_t|Y_0)$, we obtain:

$$\iint P_0(Y_0) P_t(Y_t|Y_0) \langle s_\theta(Y_t), \nabla_{Y_t} \log P_t(Y_t|Y_0) \rangle dY_0 dY_t. \quad (2.51)$$

This coincides with the expected value with respect to the variable $P_0(Y_0)P_t(Y_t|Y_0)$, that is, the expected value over the joint distribution $P_t(Y_t, Y_0)$, namely:

$$\mathbb{E}_{q(Y_0, Y_t)} [\langle s_\theta(Y_t), \nabla_{Y_t} \log q(Y_t|Y_0) \rangle] \quad (2.52)$$

Applying, moreover, the same steps to the term: $\mathbb{E}_{P_t(Y_t)} \left[\frac{1}{2} \|s_\theta(Y_t)\|^2 \right]$, whose integral form coincides with: $\int q(Y_t) \left(\frac{1}{2} \|s_\theta(Y_t)\|^2 \right) dY_t$, which, exploiting Bayes' property on $P_t(Y_t)$ conditioned on Y_0 , we rewrite as: $= \iint P_t(Y_0, Y_t) \left(\frac{1}{2} \|s_\theta(Y_t)\|^2 \right) dY_0 dY_t$,

which coincides with: $= \mathbb{E}_{P_t(Y_0, Y_t)} \left[\frac{1}{2} \|s_\theta(Y_t)\|^2 \right]$, which allows us to write the loss, recombining the norm, as:

$$\mathbb{E}_{P_t(Y_0, Y_t)} \left[\frac{1}{2} \|s_\theta(Y_t) - \nabla_{Y_t} \log P_t(Y_t|Y_0)\|^2 \right] \quad (2.53)$$

Moreover, the term $\nabla_{Y_t} \log P_t(Y_t|Y_0)$, recalling from (2.26), coincides with: $\frac{(Y_t - Y_0 e^{-t})}{\sqrt{1 - e^{-2t}}}$, so we can rewrite (2.45) as:

$$\int_0^T dt \mathbb{E}_{P_t(Y_t, Y_0)} \left\| s_\theta(Y_t) - \frac{(Y_t - Y_0 e^{-t})}{1 - e^{-2t}} \right\|^2 \quad (2.54)$$

Remark 2.2.1. : The term $\mathbb{E}_{P_t(Y_t, Y_0)} \left[\left\| s_\theta(Y_t) - \frac{(Y_t - Y_0 e^{-t})}{1 - e^{-2t}} \right\|^2 \right]$, i.e.: $\mathbb{E}_{P_t(Y_t|Y_0)P(Y_0)} \left[\left\| s_\theta(Y_t) - \frac{(Y_t - Y_0 e^{-t})}{1 - e^{-2t}} \right\|^2 \right]$. However, once Y_0 is fixed, since $Y_t = Y_0 e^{-t} + \sqrt{1 - e^{-2t}} \epsilon$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, it then becomes a random variable depending only on $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and so the expectation becomes:

$$\mathcal{L}_{DSM}(\theta) = \mathbb{E}_{Y_0 \sim q(Y_0)} \left[\mathbb{E}_\epsilon \left[\left\| s_\theta(Y_0 e^{-t} + 1 - e^{-2t} \epsilon, t) + \frac{\epsilon}{1 - e^{-2t}} \right\|^2 \right] \right]$$

However, we only have a limited sample of $Y_0 = \{x_1, \dots, x_N\}$, so we can restrict ourselves to producing a Crude Monte Carlo (**CM**) estimate of it, and since we inject the same noise into all the samples, $\epsilon = \epsilon_j \forall j = 1, \dots, N$, we obtain that at time t :

$$\mathcal{L}_t \approx \frac{1}{N} \sum_{i=1}^N \left\| s_\theta(Y_{t,i}) - \frac{(Y_{t,i} - x_i e^{-t})}{1 - e^{-2t}} \right\|^2 \quad (2.55)$$

Therefore, building upon the previous calculations and as anticipated in (2.2.1), we can reparameterize the Denoising Score Matching objective to express the loss directly in terms of the noise rather than the score.

Since the corrupted state at time t is defined as $Y_t = Y_0 e^{-t} + \sqrt{1 - e^{-2t}} \epsilon$, we can redefine the output of our neural network to directly predict the noise component ϵ . Indeed, by substituting $s_\theta(Y_t, t) = -\frac{1}{\sqrt{1 - e^{-2t}}} \epsilon_\theta(Y_t, t)$ into the expected value calculated in the last step, the normalization constants factor out, and the term inside the norm simplifies.

In this way, minimizing the distance between the scores reduces to minimizing the simple squared vector difference between the noise predicted by the model, $\epsilon_\theta(Y_t, t)$, and the actual injected noise, ϵ . By substituting this equivalence and integrating the squared error over the entire time domain, we obtain the final formulation of the loss:

$$\mathcal{L}(\theta) = \int_0^T dt \lambda(t) \mathbb{E}_{Y_0 \sim q(Y_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\|\epsilon_\theta(Y_t, t) - \epsilon\|^2 \right] \quad (2.56)$$

This elegant formulation highlights how the entire training of diffusion models essentially reduces, in practice, to optimizing a denoising network tasked with progressively estimating and subtracting Gaussian noise from the data.

Chapter 3

Learning Dynamics of Diffusion Models

In this second part we will focus on a deep interpretation that can be obtained from a detailed analysis of the *Backward* process. In particular we will analyze the results obtained in the papers by Biroli et al. (2024) and Biroli and Mezard (2023), also carefully analyzing the assumed hypotheses and their practical implications. Indeed, as already intuited by the authors themselves in Bonnaire et al. (2025), modeling the very network tasked with estimating the *Score Function* requires careful attention in the choice of parameters, architecture, training duration, and more, all of which depend absolutely on the type of problem we are analyzing, and in particular on its effective dimension d_{eff} ; indeed, it is well known that many problems, such as images, do not cover the entire spatial dimension ($\mathbb{R}^d$, or in the case of images $[0, 255]^{3 \times d}$), but rather occupy a *Manifold*, that is, a lower-dimensional submanifold. We can therefore say, broadly speaking, that the effectiveness of the modeling is conditioned by the following induced biases:

- **Sample finiteness:** The availability of a finite sample of N data points, embedded in a space of dimension d , defines an intrinsic limit to the estimation, making it strictly dependent on the density of the available observations.
- **Network finiteness:** The choice of architecture plays a determining role; ideally the **Score Function** would belong to the class of functions approximable by the network, but the finite nature of the parametric layers does not at all guarantee such inclusion.
- **Search algorithm:** Even the space of functions actually representable by the architecture undergoes an operational restriction due to the optimizer, which does not explore its entire extent, introducing further limits to our modeling.

These aspects will be analyzed further later on, but in an approximate way, based on personal, limited knowledge of the recent literature on the subject. However, a deep understanding of these phenomena is still under development, being a topic of very recent research.

3.1 Learning of the Covariance matrix in the Backward Process

Given the above, an effective method for analyzing the understanding of the distribution obtained as the *train samples* $\{x_1, \dots, x_N\}$ vary consists in isolating the contribution of the sampling bias. Let us therefore suppose we are operating in an ideal regime, not subject to limitations due to the network's architecture, with the goal of understanding how the **score**, i.e. the term we estimate — in this case therefore exactly — of the reverse SDE (A.10) is learned.

Starting from the analysis carried out by Biroli and Mezard (2023), and considering $P_t(x)$ defined in (2.30), the authors, for a more explanatory treatment, consider the real data distribution $P_0(y_0)$ as a Gaussian $\mathcal{N}(0, C^0)$ built from data distributed according to $P_0(y_0) = \frac{1}{N} \sum_{i=1}^N \delta(y_0 - x_i)$, whose covariance matrix is given by:

$$D_{ij}^t = \frac{1}{N} \sum_{\mu=1}^N x_i^\mu(t) x_j^\mu(t). \quad (3.1)$$

Analyzing the deviation between D^0 and C^0 will allow us to quantify the error caused by **sample finiteness**. Since, as stated, we are interested in how, in this circumstance, we approximate the Score term, we then consider the quantity: $\nabla_x \log P_t(x) = -(C^t)^{-1}x = -K^t x$, where K^t is defined as:

$$K^t = \frac{K^0}{e^{-2t} + \Delta_t K^0}. \quad (3.2)$$

Considering the matrix D^t , we can decompose each element as its exact expected value, i.e. C^t , plus a statistical fluctuation:

$$D_{i,j}^t = C_{i,j}^t + \delta C_{i,j}^t. \quad (3.3)$$

For the reasons stated in (2), we can rewrite the empirical data matrix as:

$$D^t = C^t + \frac{1}{d^{a/2}} R^t = C^t + \mathcal{O}(d^{-a/2}) R^t, \quad (3.4)$$

with $N = d^{a+1}$. We are now interested in understanding the spectral structure of D^t as the parameter a varies. In particular we can distinguish the following regimes:

- **Oversampling regime** ($N \gg d^2$): If we consider $a > 1$ (from which we obtain $N \gg d^2$), substituting into (B.10) we can make explicit the order of magnitude of the noise term. In the following case, the intensity of the perturbation is strictly smaller than the average spacing between the eigenvalues of C^t , which scales as $\mathcal{O}(d^{-1})$. As a consequence, the eigenvalues undergo a maximum shift of order $\mathcal{O}(d^{-a/2}) \ll \mathcal{O}(d^{-1})$, making the empirical fluctuations entirely negligible. The eigenvectors are modified only in a sub-dominant way and we can assume $D^t \approx C^t$.
- **Intermediate regime** ($d \ll N \ll d^2$): If we consider $0 < a \leq 1$ (from which we obtain $d \ll N \ll d^2$). In this case the intensity of the perturbation is now

strictly greater than the average spacing between the eigenvalues of C^t , which scales as $\mathcal{O}(d^{-1})$. As a consequence, the eigenvalues undergo a shift of order $\mathcal{O}(d^{-a/2}) \gg \mathcal{O}(d^{-1})$, exceeding their typical distance. This overlap causes strong *scrambling* of the eigenvectors, which lose alignment with the exact directions of the thermodynamic limit, making the sampling perturbation structurally relevant. Nevertheless, it can be shown (intuitively in (3), while for a rigorous treatment we refer to Biroli and Mezard (2023)) that in this regime the eigenvalue density can still be reproduced faithfully.

- **Strong undersampling regime** ($N \lesssim d$): This scenario corresponds to the case where the scale parameter satisfies $a \leq 0$ (recalling the parametrization $N \propto d^{a+1}$). In this extreme situation, the intensity of the perturbation introduced by the sample noise matrix ceases to be a sub-dominant term. The remainder term, which scales as $\mathcal{O}(d^{-a/2})$, becomes of the same order of magnitude as the original signal or dominates it completely. As a consequence, the perturbation catastrophically alters not only the eigenvectors (which lose all alignment with the exact directions), but also irreparably deforms the eigenvalue density.

It is essential to observe that in this formulation we have assumed that $P_0(Y_0)$ is Gaussian-distributed, which is in general false, since $P_0(Y_0)$ is precisely the true continuous distribution that we want to estimate starting from the finite sample τ , and moreover we have assumed the **Exact Empirical Score Hypothesis**, which means that we estimate exactly the Score function. However, in practice it is useful to consider the error generally committed both by sampling and by the class $\mathcal{G}$ of functions approximable by our network. In general, for this type of analysis one resorts to the definition of **Generalization Risk** defined as:

$$\ell(g(\cdot | \hat{\theta}_n)) = \ell(g(\cdot | \theta^*)) + \underbrace{[\ell(g(\cdot | \theta_n^*)) - \ell(g(\cdot | \theta^*))]}_{\text{approximation error}} + \underbrace{[\ell(g(\cdot | \hat{\theta}_n)) - \ell(g(\cdot | \theta_n^*))]}_{\text{statistical error}}. \quad (3.5)$$

However, this formulation too rests on several assumptions, such as assuming that the network is actually able to find the optimum within the class of approximating functions; all these assumptions will be analyzed in greater detail in (3.3).

3.2 Learning Dynamics and Phase Transitions of the Generative Process

In this second analysis we will address another of the conceptual and theoretical limits that emerge in the analysis of the dynamics of the generative process as a function of some parameters already introduced previously, such as N , i.e. the number of data points, and d , the dimension in which they live. As before, we will set aside the role played by the network, assuming that it is able to faithfully approximate the Score function (*Exact Empirical Score Hypothesis*). In particular we will see how the Langevin dynamics associated with the *backward* process thus encounters, in the thermodynamic limit, two phase transitions: a first one that we can interpret as a symmetry breaking in which the generative dynamics of the data converges

toward a class, and a second transition that corresponds to an irreversible collapse of the configurations it can assume within the class, zeroing out the fluctuations and thus collapsing onto the training data itself. Colloquially we can define this second transition as perfect memorization, since it paradoxically coincides with the total loss of information and generalization ability. We therefore want to formalize analytically what has been said, interpreting the dynamics of this process through statistical mechanics, arriving at showing how the boundary between true learning and sterile memorization is, to all effects, governed by rigorous phase transitions.

3.2.1 Regimes and phase transitions of the Backward process

As elegantly argued by Biroli et al. (2024), under the hypotheses described above the authors show that the backward generative trajectory passes through three distinct thermodynamic regimes. **Regime I** is the noise-dominated stage: at these time scales the information is completely inaccessible and the dynamics reduces to a pure, chaotic Brownian motion. However, as the noise level is lowered, the system undergoes a first phase transition at the so-called **speciation time** (t_s), that is, a spontaneous symmetry breaking that causes the trajectory to specialize toward a specific category. This transition opens the window onto a second thermodynamic phase called the **Generalization Regime**, in which the model shows itself capable of learning and generalizing. However, if the process is pushed beyond a second critical point, identified as the **collapse time** (t_c), a glassy phase transition is triggered that drags the system into the **Memorization Regime**. In this terminal phase, we can say that in a certain sense entropy runs out; within the *quenched* system, the *annealed* approximation no longer holds, and within the class of a specific dataset element, we collapse onto the element itself. The generative capacity thus remains compromised, leaving the system frozen in a sterile state of absolute memorization. As we will now analyze in detail, the entire effectiveness of diffusion models lies precisely in the delicate thermodynamic balance governed by these phase transitions.

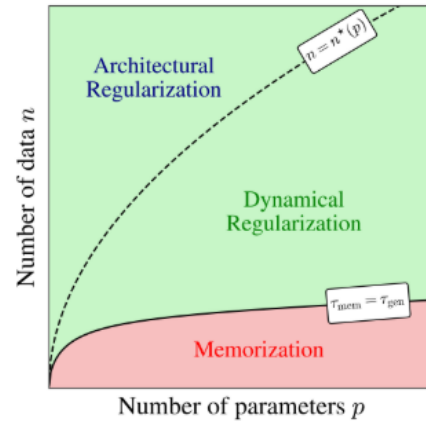


Figure 3.1. In the image we can distinguish the three regimes: the memorization regime, the generalization regime, i.e. the *dynamical regime*, and the purely noisy one, i.e. the *architectural bias regime*. Here we see the regimes as if they were also an evolution for the network, but recall that our analyses take place under the *Exact Empirical Score* hypothesis.

3.2.2 Speciation time

As introduced, we can interpret this as the instant at which the first of the two phase transitions of the *backward* process occurs. In particular, in this first analysis we will consider a **symmetry breaking**, or bifurcation in the non-symmetric case,

of the potential associated with the dynamics. Here too we assume that the network finds the absolute minimum, and thus we effectively have an **exact approximation** of the score function. Considering a simplified treatment of the problem, as done in Biroli et al. (2024), suppose we have two Gaussians centered at $+m$ and $-m$; then (2.30) becomes:

$$P_t(Y_t) = \frac{1}{2(2\pi\Gamma_t)^{N/2}} \left[\exp\left(-\frac{\|Y_t - \mathbf{m}e^{-t}\|^2}{2\Gamma_t}\right) + \exp\left(-\frac{\|Y_t + \mathbf{m}e^{-t}\|^2}{2\Gamma_t}\right) \right] \quad (3.6)$$

We observe that $\text{Cov}(Y_t) = \Gamma_t$, considering X thus sampled from a cluster with mean $\mathbf{m}$ and covariance matrix $\sigma^2 I$, and since

$$Y_t = e^{-t}X + Z_t$$

where $Z_t \sim \mathcal{N}(0, \Delta_t I)$ is noise independent of X , with $\Delta_t = 1 - e^{-2t}$; then, exploiting the linearity of covariance for independent variables, we obtain: $\text{Cov}(Y_t) = \text{Cov}(e^{-t}X + Z_t) = \text{Cov}(e^{-t}X) + \text{Cov}(Z_t)$, and thus:

$$\Gamma_t = \sigma^2 e^{-2t} + \Delta_t \quad (3.7)$$

Computing the logarithmic gradient $\nabla_{Y_t} \log P_t(Y_t)$ and expanding the squares of the norms, we obtain the fraction $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ (with $x = \mathbf{m}^T Y_t \frac{e^{-t}}{\Gamma_t}$), which coincides with $\tanh(x)$, from which we can deduce that:

$$s(Y_t, t) = -\frac{Y_t}{\Gamma_t} + \mathbf{m} \frac{e^{-t}}{\Gamma_t} \tanh\left(\mathbf{m}^T Y_t \frac{e^{-t}}{\Gamma_t}\right) \quad (3.8)$$

To understand whether the trajectory is heading toward $+\mathbf{m}$ or $-\mathbf{m}$, we do not need to look at all N coordinates of Y_t (one is enough, as stated). It is thus enough to look at its projection $q(t)$ onto the axis of the means:

$$q(t) = \frac{1}{\sqrt{N}} \mathbf{m}^T Y_t \quad (3.9)$$

From here, if we consider $\frac{\partial V}{\partial q} = -q - 2s_q(q)$, we furthermore obtain:

$$V(q, t) = \left(\frac{2}{\Gamma_t} - 1\right) \frac{q^2}{2} - 2\mu^2 \log \cosh\left(q\sqrt{N} \frac{e^{-t}}{\Gamma_t}\right) \quad (3.10)$$

where $\mu^2 = \frac{\|\mathbf{m}\|^2}{N}$, i.e. the signal intensity. Since we can interpret the Speciation time as the instant at which we transition from a single-well system to a two-well (or, in general, n -well) system, we then consider:

$$\left. \frac{\partial^2 V}{\partial q^2} \right|_{q=0, t=t^*} = 0 \quad (3.11)$$

From which we obtain:

$$\frac{\partial^2 V}{\partial q^2} = 1 - 2\mu^2 \left(e^{-t}\sqrt{N}\right)^2 \frac{1}{\cosh^2(qe^{-t}\sqrt{N})} \quad (3.12)$$

which, considering $\left. \frac{\partial^2 V}{\partial q^2} \right|_{q=0} = \left(\frac{2}{\Gamma_t} - 1 \right) - 2\mu^2 N \frac{e^{-2t}}{\Gamma_t^2} = 0$, and thus $e^{-4t} + 2\mu^2 N e^{-2t} - 1 = 0$, leads us to $t^* = -\frac{1}{2} \log \left(\sqrt{\mu^4 N^2 + 1} - \mu^2 N \right)$, which can be approximated, giving the result for t^* :

$$t^* = \frac{1}{2} \log(2N\mu^2) \quad (3.13)$$

Reformulation of the transition via mean field

We apply the deterministic change of variable $\tilde{Y}_t := e^t Y_t$, which absorbs the linear drift of the OU-VP SDE. From the VP transition probability $Y_t | Y_0 \sim \mathcal{N}(Y_0 e^{-t}, \Delta_t)$ it follows immediately, setting $\sigma_t^2 := e^{2t} \Delta_t$:

$$\tilde{Y}_t | Y_0 \sim \mathcal{N}(Y_0, \sigma_t^2), \quad \sigma_t^2 = e^{2t} \Delta_t = e^{2t} - 1 \quad (3.14)$$

that is, a genuine Variance Exploding-type diffusion: the signal Y_0 no longer decays, while the variance σ_t^2 diverges as $t \rightarrow \infty$ (hence the name).

In the limit $\sigma \rightarrow 0$ for $P_0(Y_0) = \frac{1}{2}(\delta(Y_0 - m) + \delta(Y_0 + m))$, the marginal becomes:

$$P_t(\tilde{Y}_t) = \frac{1}{2\sqrt{2\pi\sigma_t^2}} \left[\exp\left(-\frac{(\tilde{y}_t - m)^2}{2\sigma_t^2}\right) + \exp\left(-\frac{(\tilde{y}_t + m)^2}{2\sigma_t^2}\right) \right] \quad (3.15)$$

Applying Bayes' Theorem we obtain the posterior:

$$P_t(Y_0 = m | \tilde{Y}_t) = \frac{\exp\left(\frac{m}{\sigma_t^2} \tilde{y}_t\right)}{\exp\left(\frac{m}{\sigma_t^2} \tilde{y}_t\right) + \exp\left(-\frac{m}{\sigma_t^2} \tilde{y}_t\right)} \quad (3.16)$$

and equivalently for $-m$, from which the conditional expectation follows:

$$\langle Y_0 \rangle_{\tilde{Y}_t} = m \tanh\left(\frac{m}{\sigma_t^2} \tilde{Y}_t\right) \quad (3.17)$$

Recalling now Tweedie's Formula in the genuine VE case,

$$\nabla_{\tilde{Y}_t} \log P_t(\tilde{Y}_t) = -\frac{\tilde{Y}_t - \langle Y_0 \rangle_{\tilde{Y}_t}}{\sigma_t^2} \quad (3.18)$$

we isolate the conditional expectation:

$$\langle Y_0 \rangle_{\tilde{Y}_t} = \tilde{Y}_t + \sigma_t^2 \nabla_{\tilde{Y}_t} \log P_t(\tilde{Y}_t) \quad (3.19)$$

Imposing $\nabla_{\tilde{Y}_t} \log P_t(\tilde{Y}_t) = 0$:

$$\tilde{Y}_t = \langle Y_0 \rangle_{\tilde{Y}_t} \quad (3.20)$$

which, substituted into (3.17), gives the self-consistency equation:

$$\langle Y_0 \rangle_{\tilde{Y}_t} = m \tanh\left(\frac{m}{\sigma_t^2} \langle Y_0 \rangle_{\tilde{Y}_t}\right) \quad (3.21)$$

Dividing by m and defining $M(t) = \langle Y_0 \rangle_{\tilde{Y}_t} / m$, we obtain the Mean Field equation in the VE formulation:

$$M(t) = \tanh(\tilde{\lambda}(t)M(t)), \quad \tilde{\lambda}(t) = \frac{m^2}{\sigma_t^2} = \frac{m^2}{e^{2t} - 1} \quad (3.22)$$

Consistency of the critical time. Expanding $\tanh(x) \approx x - \frac{1}{3}x^3$ as in the VP case, the transition occurs at $\tilde{\lambda}(t_s) = 1$. Setting $m^2 = N\mu^2$:

$$\frac{N\mu^2}{e^{2t_s} - 1} = 1 \implies e^{2t_s} = N\mu^2 + 1 \quad (3.23)$$

$$t_s = \frac{1}{2} \log(N\mu^2 + 1) \quad (3.24)$$

Remark In Biroli et al. (2024), to find the speciation time we considered: $\frac{\partial V}{\partial q} = -q - 2s_q(q)$. If we were to consider its VE formulation, we would instead obtain $-\frac{2}{e^t} s'_q(0)$, and therefore we would get $s'_q(0) = 0$ (used in the previous proof) rather than $s'_q(0) = -\frac{1}{2}$. Had we considered the equivalent VE form with two Gaussians, we would have obtained:

$$\frac{1}{\Gamma_t} - N\mu^2 \frac{e^{-2t}}{\Gamma_t^2} = 0$$

from which, writing Γ_t explicitly:

$$t^*(\sigma) = \frac{1}{2} \log(N\mu^2 - \sigma^2 + 1) \quad (3.25)$$

In this case, sending $\sigma \rightarrow 0$ we recover exactly the result obtained via mean field (3.24), assuming in both cases that the dynamics can be projected onto a space $d \ll N$. In the mean-field treatment, then, we deal with Dirac deltas, but unlike what was introduced in (2.10), this time we do not consider deltas centered on the data, but rather it is as if we assumed a binary output of these.

This analysis raises some questions, since in the VP formulation, in order to identify a bifurcation — i.e. the instant at which the dynamics commits to specializing — we found something dependent on the scheduler $\beta(t) = 2$. This, as we can see, anticipated the moment at which it might make more sense to specialize, namely the one at which the distribution turns multimodal, as dictated by the VE formulation.

Comparison with the Variance Exploding formulation (Ambrogioni 2023)

The proposed approach is inspired by Ambrogioni (2023), where, however, the Variance Exploding (VE) formulation of an Ornstein–Uhlenbeck process is considered (i.e., a pure Brownian motion). In particular, compared to the analogous formulation introduced by us with $\beta(t) = 2$, they define $\beta(t) = (t\sigma^2)^{-1}$, and in the mean-field limit ($N \rightarrow \infty$) they obtain the self-consistency equation:

$$m = \tanh\left(\frac{m}{t\sigma^2}\right) \quad (3.26)$$

By analyzing the neighborhood of $m = 0$ through a Taylor series expansion, the same phase transition associated with the critical time is recovered:

$$t_c \propto \frac{1}{\sigma^2} \quad (3.27)$$

3.2.3 Collapse time

As anticipated, it is possible to interpret the *backward* process through a second phase transition, known as the **collapse time**, which we will denote by t_c . This transition describes the passage from the *generalization* regime to the *collapse* regime. The analytical treatment that follows takes up what was introduced in Biroli et al. (2024).

Assuming, as before, that the network exactly reproduces the score function associated with the distribution throughout the whole temporal dynamics, we can interpret this phenomenon, in the language of statistical mechanics, as the passage from a phase in which the partition function associated with the class we are analyzing is dominated by many possible energetic configurations, to a phase in which only one becomes possible; in particular, this single one is the one that minimizes the energy for the partition function. Moreover, given how diffusion models are defined, during the generative process the variance of the distribution is progressively reduced, collapsing as $t \rightarrow 0$ toward a Dirac delta centered on the data, as discussed in (2.2.1). From the point of view of statistical mechanics, the collapse of the variance and the minimization of the energy are closely interconnected phenomena: the reduction of stochastic noise is mathematically equivalent to a progressive lowering of the system's temperature. In this regime, under suitable isotropy hypotheses on the noise, the diffusion dynamics is rigorously configured as the minimization of the free energy in the space of distributions, a topic that will be explored further in (3.3). However, in high-dimensional contexts (*Curse of dimensionality*), the predominance of the entropic term makes the exploration and effective collapse toward the energy minima a computationally delicate process.

In particular, this represents a passage from the *annealed* approximation in *quenched* phase, where the system can explore the entire state space, to a phase in which the system remains constrained to specific energy levels. The description of such systems in statistical physics typically occurs by means of the so-called **Random Energy Models** (see Schehr and Zamponi (s.d.) for further details), while more rigorous treatments can instead be formulated through the replica method Parisi (1979).

Let us therefore consider (2.10), from which we obtain in particular:

$$P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \frac{1}{(2\pi\Delta_t)^{D/2}} e^{-E_i} \quad (3.28)$$

where $E_i = \frac{1}{2\Delta_t} \|Y_t - x_i e^{-t}\|^2$, and in particular, if we consider the normalization constant $Z = \sum_{i=1}^N Z_i = \sum_{i=1}^N e^{-E_i}$, we can rewrite the marginal distribution as:

$$P_t(Y_t) = \frac{Z_1 + Z_{2\dots N}}{(2\pi\Delta_t)^{d/2}}, \quad (3.29)$$

If we assume that Y_t was generated starting from x_1 , then in general $Y_t = x_1 e^{-t} + z\sqrt{\Delta_t}$ holds, from which we obtain that $Z_1 = e^{-\frac{1}{2}\|z\|^2}$, while if we consider:

$$Z_{2\dots N} = \sum_{\mu=2}^N e^{-E_\mu} \quad (3.30)$$

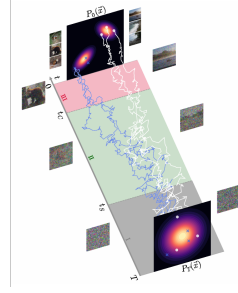


Figure 3.2. Visualization of the two phase transitions

i.e. the partition function of the other data points in the dataset, with the energy levels defined as:

$$E_{x_i} = \frac{1}{2\Delta_t} \left\| (x_1 - x_i)e^{-t} + z\sqrt{\Delta_t} \right\|^2. \quad (3.31)$$

which, by expanding the square, we can write as:

$$E_{x_i} = \frac{e^{-2t}}{2\Delta_t} \|x_1 - x_i\|^2 + \frac{1}{2} \|z\|^2 + \frac{e^{-t}}{\sqrt{\Delta_t}} (x_1 - x_i) \cdot z. \quad (3.32)$$

In the limit $d \rightarrow \infty$, we have that:

$$\|x_1 - x_i\|^2 \simeq 2d\sigma^2, \|z\|^2 \simeq d, \quad (3.33)$$

by the concentration of measure treated in (1), it follows immediately that:

$$Z_1 \simeq e^{-d/2}. \quad (3.34)$$

On the other hand, if we want to compute the expected value of (3.30), since these are all i.i.d. variables, we obtain:

$$\mathbb{E}[Z_{2\dots N}] = (N-1) \mathbb{E}[e^{-E_{x_i}}], \quad (3.35)$$

We now assume that the variables come in general from two clusters, one centered at m and the other at $-m$, and thus $N = n_+ + n_-$. This allows us to rewrite $Z_{2\dots N}$ as:

$$Z_+ = \sum_{i=2}^{n_+} e^{-E_{x_i}}, \quad Z_- = \sum_{i=n_++1}^{n_-} e^{-E_{x_i}}. \quad (3.36)$$

In particular, we assume that x_1 comes from the distribution centered at m . However, in this case we have a sum of exponential variables ($N = e^{\alpha d}$), and not a single variable as in Z_1 . In particular, since we are considering the thermodynamic limit, i.e. $E \rightarrow \infty$ as $d \rightarrow \infty$, we introduce an intensive quantity such as $\varepsilon_i = \frac{E_i}{d}$, from which we obtain that $Z_+ = \sum_{i=1}^{n_+} e^{-d\varepsilon_i}$. However, it is evident that if we consider $\mathbb{E}[e^{-d\cdot\lambda\varepsilon}]$, not all the terms of this sum affect the final calculation in the same way, since we are averaging over a nonlinear variable. If we then go on to analyze the distribution of the ε 's, these do not fluctuate around the mean at order $\mathcal{O}(\sqrt{d})$, and we cannot assume the validity of the CLT and of the concentration-of-measure theory. We must instead resort to **Large Deviations Theory**, and we assume that the probability density $\rho(\varepsilon)$ of observing one of these macroscopic fluctuations ε decays exponentially fast as the dimension d increases: $\rho(\varepsilon) \approx e^{-d\cdot f(\varepsilon)}$. We therefore want to determine the rate function $f_t(\varepsilon)$, but computing it directly from the definition of probability requires computing terrifying volumes of integrals in d dimensions, taking into account complex geometric constraints; we therefore consider:

$$g_t(\lambda) := \lim_{d \rightarrow \infty} \frac{1}{d} \log \mathbb{E}_{x_i} \left[e^{-\lambda E_{x_i}} \right] = \lim_{d \rightarrow \infty} \frac{1}{d} \log \mathbb{E} \left[e^{-\frac{\lambda}{2\Delta_t} \|(x_1 - x_i)e^{-t} + z\sqrt{\Delta_t}\|^2} \right], \quad (3.37)$$

that is, the **scaled cumulant generating function**. Since $x_1 - x_i$ and z are independent Gaussian variables, component by component, the right-hand side of (3.37) can be computed in closed form as a convolution of two Gaussians (see also

the direct computation in (1), which provides the concentration $\|x_1 - x_i\|^2 \simeq 2d\sigma^2$ used in this calculation), obtaining:

$$g_t(\lambda) = \frac{1}{2} \log \frac{\Delta_t}{\Delta_t + \lambda\sigma_t^2} - \frac{1}{2} \frac{\lambda(\Delta_t + \sigma_t^2)}{\Delta_t + \lambda\sigma_t^2}. \quad (3.38)$$

Since $g_t(\lambda)$ is a convex, differentiable function, the density of ε satisfies a large-deviations principle:

$$\rho_t(\varepsilon) \asymp e^{-d f_t(\varepsilon)}, \quad f_t(\varepsilon) = \sup_{\lambda} [-\lambda\varepsilon - g_t(\lambda)], \quad (3.39)$$

where f_t and g_t are the Legendre transform of one another: $g_t(\lambda) = \sup_{\varepsilon} [-\lambda\varepsilon - f_t(\varepsilon)]$. The stationarity point $\lambda^*(\varepsilon)$ of the extremum defining f_t is determined by:

$$\frac{\partial}{\partial \lambda} [-\lambda\varepsilon - g_t(\lambda)] = 0 \iff \varepsilon = -g'_t(\lambda^*), \quad (3.40)$$

from which, by inverting,

$$\varepsilon^*(\lambda) = -\frac{dg_t}{d\lambda} = \frac{\Delta_t^2 + 2\Delta_t\sigma_t^2 + \lambda\sigma_t^4}{2(\Delta_t + \lambda\sigma_t^2)^2}, \quad (3.41)$$

and substituting the $\lambda^*(\varepsilon)$ so obtained into $f_t(\varepsilon) = -\lambda^*\varepsilon - g_t(\lambda^*)$ yields the explicit expression:

$$\begin{aligned} f_t(\varepsilon) &= \frac{\Delta_t}{2\sigma_t^2(\sigma_t^2 + A)} \left[-8\Delta_t\varepsilon + (1 - 6\varepsilon)\sigma_t^2 + (1 + 2\varepsilon)A \right] \\ &\quad - \frac{1}{2} \log \frac{4\Delta_t\varepsilon}{\sigma_t^2 + A}, \\ A &= \sqrt{\sigma_t^4 + 8\varepsilon\Delta_t(\Delta_t + \sigma_t^2)}. \end{aligned} \quad (3.42)$$

At this point, if we want to treat the quantity $Z_+ = \sum_{\mu=2}^{n_+} e^{-d\varepsilon^\mu}$ ((3.36)), since we have characterized $\rho_t(\varepsilon)$, we can write:

$$Z_+ = n_+ \int d\varepsilon e^{-d(\varepsilon + f_t(\varepsilon))}.$$

By Laplace's method, this integral is dominated by the point ε^* at which the exponent $\varepsilon + f_t(\varepsilon)$ is minimal:

$$Z_+ \approx n_+ e^{-d \min_{\varepsilon} [\varepsilon + f_t(\varepsilon)]}, \quad (3.43)$$

from which, taking $\frac{1}{d} \log$ of both sides and using $n_+ = e^{\alpha d}$:

$$\psi_+ = \lim_{d \rightarrow \infty} \frac{1}{d} \log Z_+ = \alpha - \min_{\varepsilon} [\varepsilon + f_t(\varepsilon)]. \quad (3.44)$$

The stationarity condition for the minimum is:

$$\left. \frac{d}{d\varepsilon} [\varepsilon + f_t(\varepsilon)] \right|_{\varepsilon=\varepsilon^*} = 1 + f'_t(\varepsilon^*) = 0 \implies f'_t(\varepsilon^*) = -1.$$

From the duality relation $\lambda^*(\varepsilon) = -f'_t(\varepsilon)$ (a direct consequence of the Legendre-Fenchel transform, derivable by computing $\frac{d}{d\varepsilon} f_t(\varepsilon) = \frac{d}{d\varepsilon} [-\lambda^*(\varepsilon)\varepsilon - g_t(\lambda^*(\varepsilon))] = -\lambda^*(\varepsilon) - \frac{d\lambda^*}{d\varepsilon} [\varepsilon + \underbrace{g'_t(\lambda^*)}_{=0}] = -\lambda^*(\varepsilon)$), we obtain $\lambda^*(\varepsilon^*) = 1$. Substituting into (3.41):

$$\varepsilon^*(1) = \frac{\Delta_t^2 + 2\Delta_t\sigma_t^2 + \sigma_t^4}{2(\Delta_t + \sigma_t^2)^2} = \frac{(\Delta_t + \sigma_t^2)^2}{2(\Delta_t + \sigma_t^2)^2} = \frac{1}{2}.$$

Substituting $\lambda^* = 1$ and $\varepsilon^* = \frac{1}{2}$ into (3.44):

$$\begin{aligned} \psi_+ &= \alpha - [\varepsilon^* + (-\lambda^*\varepsilon^* - g_t(\lambda^*))] \Big|_{f_t(\varepsilon^*) = -\lambda^*\varepsilon^* - g_t(\lambda^*)} \\ &= \alpha - \varepsilon^* - f_t(\varepsilon^*) = \alpha + g_t(1) \\ &= \alpha + \frac{1}{2} \log \frac{\Delta_t}{\Delta_t + \sigma_t^2} - \frac{1}{2}. \end{aligned} \quad (3.45)$$

where we substituted the value of $g(1)$ starting from (3.37). The preceding **annealed** approximation is valid in the **quenched** regime, since, observing the definition of ψ_+ , we substituted into (3.44) with $n_+ = e^{\alpha d}$. However, given how it was introduced, once all the variables concentrate at ε^* , i.e. the minimizer of Z_+ and hence in general the state ε that minimizes the free energy (the saddle point in the complex plane), the annealed approximation doesn't work anymore. In this case then $\mathcal{N}(\varepsilon^*) = n_+$, i.e. all the variables concentrate at ε^* , and thus, if we consider (3.36), by Laplace's method we had concluded that:

$$Z_+ \approx n_+ e^{-d(\varepsilon^* + f_t(\varepsilon^*))} = \mathcal{N}(\varepsilon^*) \cdot e^{-d\varepsilon^*} = n_+ e^{-d\varepsilon^*} \quad (3.46)$$

(where we used the fact that: $\mathcal{N}(\varepsilon^*) = \underbrace{e^{-d\varepsilon^*}}_{\text{probability of a single state}} \times \underbrace{n_+}_{\text{number of states}}$). We have therefore obtained, from (3.46), that in this regime $f_t(\varepsilon^*) = 0$. It follows that:

$$\psi_+^{ann} \approx \lim_{d \rightarrow \infty} \frac{1}{d} n_+ \log(e^{-\alpha d(\varepsilon^* + f_t(\varepsilon^*))}) = \lim_{d \rightarrow \infty} \frac{1}{d} e^{\alpha d} \log(e^{-\alpha d \frac{1}{2}}) = -\frac{1}{2} \quad (3.47)$$

To find the critical value t_c at which the transition occurs, we set:

$$\psi_+^{ann}(\varepsilon) = \psi_+(\varepsilon) \quad (3.48)$$

from which we thus obtain:

$$\alpha + \frac{1}{2} \log \left(\frac{\Delta_t}{\Delta_t + \sigma_t^2 e^{-2t}} \right) - \frac{1}{2} = -\frac{1}{2} \quad (3.49)$$

and (recalling $\Delta_t = 1 - e^{-2t}$ and $\alpha = \frac{\log n}{d}$), we then obtain that:

$$t_C = \frac{1}{2} \log \left(1 + \frac{\sigma^2}{N^{2/d} - 1} \right) \quad (3.50)$$

In an entirely analogous way it is possible to treat $\psi_-(\varepsilon)$, as done in Biroli et al. (2024), provided the Legendre transform $g_t(\lambda)$ is modified, which in that case becomes:

$$g_t(\lambda) = \frac{1}{2} \log \left(\frac{\Delta_t}{\Delta_t + \lambda \sigma_t^2} \right) - \frac{1}{2} \frac{\lambda [M_t + \Delta_t + \sigma_t^2]}{\Delta_t + \lambda \sigma_t^2} \quad (3.51)$$

where $M_t = 4(\mathbf{m}^2/d)e^{-2t}$. This difference emerges because the quantity $(\mathbf{x} - \mathbf{m}e^{-t})^2/d$ concentrates, in the large- d limit, around the value $4(\mathbf{m}^2/d)e^{-2t} + \sigma_t^2 + \Delta_t$, since the data associated with Z_- are sampled from a Gaussian with mean $-\mathbf{m}$ (and not $\mathbf{m}$, where the data point x_1 , with respect to which we are estimating the distance, is located). we obtain in this case that:

$$\psi_- = \lim_{d \rightarrow \infty} \frac{1}{d} \log Z_- = \begin{cases} \alpha + \frac{1}{2} \log \frac{\Delta_t}{\Delta_t + \sigma_t^2} - \frac{1}{2} - \frac{1}{2} \frac{M_t}{\Delta_t + \sigma_t^2}, & t < t_{\text{cond}}, \\ -\frac{1}{2} \left[1 + \frac{\Delta_t M_t}{(\Delta_t + \sigma_t^2)^2} \right], & t > t_{\text{cond}}. \end{cases} \quad (3.52)$$

from which we obtain that: $t_{\text{cond}} = \frac{1}{2} \log \left[1 + \frac{\sigma^2}{n^{2/d} - 1} \right]$. However, even though the expression for the free energy varies, the computation to find the minimum and the associated condensation transition still leads to the same critical time t_c ; this is because, in general, t_c depends on the disorder of the system (how much it fluctuates around the mean). Moreover we can observe that the free energy of the opposite cluster will always be strictly smaller: $\psi_- < \psi_+$, and thus, in a high-dimensional regime ($d \rightarrow \infty$), the probabilities are compared on an exponential scale. This means that the weight of the opposite cluster ($Z_- \sim e^{-d\psi_-}$) is mathematically "crushed" and canceled out by the weight of the original cluster ($Z_+ \sim e^{-d\psi_+}$), making it irrelevant for the purposes of the dynamics. As a consequence, the critical collapse time t_c does not arise from a competition between the two clusters, but is dictated solely by an internal competition within the cluster of membership: the transition occurs at the exact moment when the energy of the whole set of data points in the cluster (Z_+) is overtaken by that of the single original target data point ($Z_1 \sim e^{-d/2}$), a condition that translates mathematically into $\psi_+(t_c) = -1/2$.

Remark. This analysis in Biroli et al. (2024) was also carried out starting from the case of two Gaussian distributions, as in the case of the **Speciation Time**. However, previously, in (3.24), we had introduced the analogous analysis through the mean-field approach, which was valid only under the assumption of the least informative prior distribution, namely the sum of Dirac delta functions located at $+m$ and $-m$. We argued that this corresponds to the case of data with binary outputs.

For this type of initial distribution, by taking the limit $\sigma \rightarrow 0$ in:

$$t_{\text{cond}} = \frac{1}{2} \log \left[1 + \frac{\sigma^2}{n^{2/d} - 1} \right] \rightarrow 0,$$

we obtain the same limiting behavior.

3.3 Implicit Regularizations of the Generative Dynamics

In this section we want to slightly widen the window of our analysis of Diffusion Models, also to cases in which some of the previously made hypotheses no longer hold, such as:

1. Exact Empirical Score Hypothesis

2. Curse of Dimensionality

and others. The goal would be to provide a fairly reliable analysis of what actually happens in practice when we use these models, and to quantify how the design of the architecture, and thus the number of parameters and so on, depends on the complexity of the problem we are treating. As already mentioned earlier, however, this is a subject of the most recent research; therefore, unfortunately, many results are fragmentary, cover specific cases, and a general theory on the matter has not yet been developed in a definitive way.

1. Curse of Dimensionality and the Manifold Hypothesis

Focusing first on the dimensionality of the data, we had mentioned how the **Manifold Hypothesis** holds, namely that in many circumstances our data live on a manifold $\mathcal{M} \subset \mathbb{R}^d$. The case of images, as stated, is one of these, in which $\dim(\mathcal{M})$ is actually rather small compared to the ambient space (it also depends on the type of images). However, in other cases, such as the Rademacher dataset (i.e. a dataset of pure noise generated by uncorrelated random variables), the dimension of the Manifold practically coincides with the ambient space. In cases where the dimension of the problem is actually lower, we are thus not subject to the problem of the **Curse of Dimensionality** (1). More precisely, the sample complexity required to accurately estimate the score does not depend on the dimension of the ambient space d , but rather on the intrinsic dimension $d_{\text{eff}} = \dim(\mathcal{M})$; it follows that the number of samples needed scales as $N \propto e^{\alpha d_{\text{eff}}}$ rather than $N \propto e^{\alpha d}$, avoiding the exponential explosion of the data.¹ An important role of the effective dimension of the problem thus occurs in the analysis of the **Expected Generalization Risk**, and in particular in the analysis of its **CM** version, which we estimate, i.e. the **Test Loss**, which we will address carefully in the next paragraph.

¹Furthermore, the shape of our Manifold turns out to be fundamental in the analysis of the architecture we want to use. As introduced by Bronstein et al. (2021), every architecture is subject to a geometric bias that we are assuming about the data. In the case of images, which we have extensively treated, we assume that they are translation-invariant. In linear algebra, it is known that translation-equivariant (or invariant) operators take the form of circulant matrices:

$$C = \begin{pmatrix} c_0 & c_{n-1} & \dots & c_2 & c_1 \\ c_1 & c_0 & \dots & c_3 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{n-1} & c_{n-2} & \dots & c_1 & c_0 \end{pmatrix}$$

Moreover, these matrices (and convolutions in general) enjoy the fundamental property of being diagonalized by the Fourier basis, and can be expressed through the factorization

$$C = F\Lambda F^*.$$

where F represents the discrete Fourier transform (DFT) matrix, F^* its conjugate transpose, and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ the diagonal matrix containing the frequency response of the operator. However, the resulting system is not completely decoupled: the presence of nonlinear layers in fact allows the different harmonics to be coupled in a non-trivial way. To analytically describe this composition of linear invariant filters and nonlinearities, such architectures can be interpreted, in the literature, as Volterra series approximators Li et al. (2022).

Generalization Risk and Double Descent

For the reasons stated, it is therefore of interest to consider a quantity that measures, on average, the error committed on each validation dataset τ' , i.e. $\mathbb{E}_{\tau'}[\ell(g_{\tau'}^{\mathcal{G}})]$, with $g \in \mathcal{G}$ the class of functions approximable by the network. As introduced in (3.1), starting from the Generalization Risk equation (3.5), we can first of all separate the irreducible error ℓ^* from the error introduced by our approximation with respect to the optimal predictor $g^*(\mathbf{X})$:

$$\ell(g_{\tau}^{\mathcal{G}}) = \mathbb{E} \left[(g_{\tau}^{\mathcal{G}}(\mathbf{X}) - Y)^2 \right] = \ell^* + \mathbb{E} \left[(g_{\tau}^{\mathcal{G}}(\mathbf{X}) - g^*(\mathbf{X}))^2 \right] \quad (3.53)$$

Defining the discrepancy $D(\mathbf{x}, \tau) := g_{\tau}^{\mathcal{G}}(\mathbf{x}) - g^*(\mathbf{x})$, we are interested in analyzing the average behavior of the model as the training dataset $\mathcal{T}$ varies. Evaluating the expectation of the square of $D(\mathbf{x}, \mathcal{T})$ for a generic input $\mathbf{x}$, we obtain the exact decomposition:

$$\begin{aligned} \mathbb{E}_{\mathcal{T}} \left[(g_{\mathcal{T}}^{\mathcal{G}}(\mathbf{x}) - g^*(\mathbf{x}))^2 \right] &= \mathbb{E}_{\mathcal{T}} [D^2(\mathbf{x}, \mathcal{T})] \\ &= (\mathbb{E}_{\mathcal{T}} D(\mathbf{x}, \mathcal{T}))^2 + \text{Var}_{\mathcal{T}} D(\mathbf{x}, \mathcal{T}) \\ &= \underbrace{(\mathbb{E}_{\mathcal{T}} g_{\mathcal{T}}^{\mathcal{G}}(\mathbf{x}) - g^*(\mathbf{x}))^2}_{\text{pointwise squared bias}} + \underbrace{\text{Var}_{\mathcal{T}} g_{\mathcal{T}}^{\mathcal{G}}(\mathbf{x})}_{\text{pointwise variance}} \end{aligned} \quad (3.54)$$

Integrating this expression over the entire distribution of the test data, we can reformulate the expected error of the model in terms of *expected bias* and *expected variance*.

However, in practice we generally have available only a single validation test set τ' of size N_{test} . Since we cannot compute the exact expectation over infinitely many datasets, we restrict ourselves to a *Crude Monte Carlo* estimate of this quantity. The error is thus evaluated empirically through the sum over the samples available to us:

$$\hat{\ell}_{\tau'}(g_{\tau}^{\mathcal{G}}) = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left(g_{\tau}^{\mathcal{G}}(\mathbf{x}_i) - y_i \right)^2 \quad (3.55)$$

We can thus reformulate the error committed by the model in the practical terms of this sample average. However, based on numerical results and the most recent studies, it has been noted that, contrary to this classical "V"-shaped interpretation — where increasing the number P of parameters decreases bias but increases variance — modern theory predicts a further second descent of this quantity in the **Overparameterized** regime.

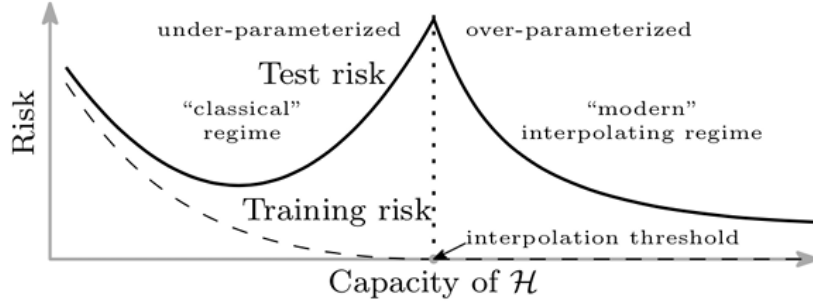


Figure 3.3. Double descent Lafon and Thomas (2024): behavior of the training and test error as the ratio between parameters and data varies. After the classic bias-variance trade-off and the error peak at the interpolation threshold (where the training error goes to zero), the test error undergoes a "second descent" in the over-parameterized regime ($\mathcal{H}$ is the equivalent of our $\mathcal{G}$, typically used for polynomial regression models).

The first to introduce this phenomenon in the case of polynomial regression were Schaeffer et al. (2023). As we will see later in (3.4.3), an attempt was made to provide practical and implementation-level evidence for it also for diffusion models, finding results consistent with the theory. We have also investigated the role played by the dimension of the problem d_{eff} in this analysis, and in particular we have observed how the "**second descent**" is evidently more marked for problems with lower dimension (3.3), (3.9). This result agrees well with the analysis carried out by Han et al. (2026), who formulated the following Theorem: In a Diffusion model, if we consider an over-parameterized network trained via Gradient Descent with *early stopping*, we have that the Generalization Risk is:

$$\frac{1}{T - \delta} \int_{\delta}^T \lambda(t) \mathbb{E} \left[\left\| \underbrace{s_{\hat{\theta}}(t, X_t)}_{\text{score estimator}} - \nabla \log p(t, X_t) \right\|^2 \right] dt = \mathcal{O} \left(N^{-\frac{2}{3+d}} \right) \quad (3.56)$$

It is thus fundamental to observe how the error scales as $N^{-\frac{2}{3+d}}$ and therefore in general decreases as N increases, and moreover the rate at which this happens depends on the effective dimension of the problem: the larger the latter, the slower the decrease of the error. To obtain this result, the Expected Risk was decomposed

into four fundamental components:

$$\begin{aligned}
& \frac{1}{c(T - T_0)} \int_{T_0}^T \mathbb{E} \left[\left\| \Pi_D(f_{\mathbf{w}(\tau)}(X_t, t)) - f_*(X_t, t) \right\|_2^2 \mathbb{I}\{\|X_t\|_2 \leq R\} \right] dt \\
& \leq \frac{1}{T - T_0} \int_{T_0}^T \mathbb{E} \left[\left\| \Pi_D(f_{\mathbf{w}(\tau)}(X_t, t)) - f_\tau^K(X_t, t) \right\|_2^2 \mathbb{I}\{\|X_t\|_2 \leq R\} \right] dt \quad (\text{Coupling}) \\
& + \frac{1}{T - T_0} \int_{T_0}^T \mathbb{E} \left[\left\| f_\tau^K(X_t, t) - \tilde{f}_\tau^K(X_t, t) \right\|_2^2 \mathbb{I}\{\|X_t\|_2 \leq R\} \right] dt \quad (\text{Label Mismatch}) \\
& + \frac{1}{T - T_0} \int_{T_0}^T \mathbb{E} \left[\left\| \tilde{f}_\tau^K(X_t, t) - f_{\mathcal{H}}(X_t, t) \right\|_2^2 \mathbb{I}\{\|X_t\|_2 \leq R\} \right] dt \quad (\text{Early Stopping}) \\
& + \frac{1}{T - T_0} \int_{T_0}^T \mathbb{E} \left[\left\| f_{\mathcal{H}}(X_t, t) - f_*(X_t, t) \right\|_2^2 \mathbb{I}\{\|X_t\|_2 \leq R\} \right] dt \quad (\text{Approximation})
\end{aligned} \tag{3.57}$$

(The authors set $\mathbb{I}\{\|X_t\|_2 \leq R\}$ for the issue introduced in (2.8) related to the VP-VE formulations; in our case, therefore, we can assume this hypothesis to hold). Let us examine the meaning of each term:

- **Approximation Error:** Measures the intrinsic limitation of the architecture. It is the distance between the best possible function within the class of functions representable by the network ($\mathcal{H}$) and the true target function f_* .
- **Early Stopping (Optimization Error):** Represents the error introduced by stopping the training (Gradient Descent) at a finite time τ (typically set via `max_iter`) in order to avoid overfitting.
- **Coupling Error:** A term linked to the *Neural Tangent Kernel* theory, quantifying the error we make by using the *GD* algorithm on a finite-dimensional network (rather than in its theoretical limit).
- **Label Mismatch (Statistical Error):** This error quantifies the penalty we pay for being able to compute the gradient (or train the model) only on an empirical, finite set of N samples.

We can thus also observe from this decomposition what we had already hinted at earlier. Indeed, classical *Statistical Learning Theory* assumes that within the class of functions approximable by the network we are always able to find the optimum. However, as shown in this analysis, we must consider two further (and perhaps other) factors due to the fact that we discretize the dynamics of Gradient Descent, and moreover that this in fact stops after a certain number of epochs on a finite-dimensional network. This, however, actually has a benign effect, which we will analyze in a more articulated way in the next paragraph.

Spectral bias

If we consider a network trained via **Gradient Descent**, as intuited in the paper by Bonnaire et al. (2025) for diffusion models, but already introduced empirically by Rahaman et al. (2019) in more generic circumstances and subsequently theoretically justified in the Neural Tangent Kernel regime by Jacot et al. (2018), the gradient

descent dynamics first learns the spectral components associated with the larger eigenvalues of the kernel, or of the estimate that we actually deal with, i.e. the *batch* or, more precisely, the *mini-batch*, which typically correspond to the low frequencies of the signal. To see this intuitively, let us consider the continuous-time dynamics of GD on a quadratic loss $f(x) = \frac{1}{2}\|F(x) - y\|^2$, linearized around the initialization via the NTK:

$$\dot{x}(t) = -J(x_0)^T (F(x_0) - y), \quad (3.58)$$

and then we obtain:

$$\dot{F}(x(t)) = -K(x_0)(F(x(t)) - y) \quad (3.59)$$

Defining the error as $e(t) = F(t) - y$, the dynamics in the NTK regime is

$$\dot{e}(t) = -Ke(t).$$

Since $K = U\Lambda U^\top$, with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $e(0) = \sum_k \alpha_k u_k$, the solution is

$$e(t) = e^{-Kt}e(0) = \sum_{k=1}^n \alpha_k e^{-\lambda_k t} u_k,$$

that is, the projection of the error along the k -th eigenvector evolves as

$$u_k^\top e(t) = e^{-\lambda_k t} u_k^\top e(0).$$

Therefore, the components associated with the larger eigenvalues decay more quickly, while those associated with the smaller eigenvalues are learned only over longer times. Thus, if we were to run training for a very long time, we would end up learning the noisiest frequencies, in the case of GD associated with a specific valley of a *local minimum*. In the case of **Stochastic Gradient Descent**, the situation worsens: the noise induced by mini-batch sampling probably allows escaping the local minimum, which would normally be an advantage, but without *early stopping* the dynamics converges toward minima associated with the noisy frequencies.

Langevin dynamics of SGD

The dynamics of SGD itself can be interpreted as a Langevin Dynamics. Following Chaudhari and Soatto (2018), we define the **Diffusion Matrix** $D(x)$: if a mini-batch is sampled with replacement, the variance of the mini-batch gradients is $\text{var}(\nabla f_\delta(x)) = \frac{D(x)}{\delta}$, where

$$D(x) = \left(\frac{1}{N} \sum_{k=1}^N \nabla f_k(x) \nabla f_k(x)^\top \right) - \nabla f(x) \nabla f(x)^\top \succeq 0. \quad (3.60)$$

The continuous-time limit of SGD is thus the following SDE Chaudhari and Soatto (2018):

$$dx(t) = -\nabla f(x) dt + \sqrt{2\beta^{-1}D(x)} dW(t), \quad (3.61)$$

where $\beta^{-1} = \frac{\eta}{2\delta}$ plays the role of an effective temperature and $W(t)$ is standard Brownian motion. It is important to observe that this SDE is already *spatially non-stationary* even for fixed η and δ , since the diffusion coefficient $D(x)$ depends

explicitly on the current position $x(t)$ in weight space: unlike a standard Langevin dynamics with constant isotropic noise, here the structure of the noise adapts to the local geometry of the loss. Moreover, the authors show that SGD minimizes the Free Energy Functional

$$\rho^{\text{ss}} = \arg \min_{\rho} \mathbb{E}_{x \sim \rho} [\Phi(x)] - \frac{\eta}{2\delta} H(\rho), \quad (3.62)$$

where $H(\rho)$ is the entropy of the distribution ρ and $\Phi(x)$ is a potential that generally depends on the architecture and the dataset. The central result is that $\Phi(x) = f(x)$ *if and only if* $D(x)$ is isotropic, i.e. $D(x) = \sigma^2 I$ with σ^2 independent of x . The deep reason for this if-and-only-if lies in optimal transport theory: in the isotropic case, the SDE (3.61) is a standard Langevin dynamics, whose associated Fokker–Planck equation selects, at each step, the distribution closest (in the Wasserstein W_2 metric) in terms of Free Energy:

$$\mathcal{F}(\rho) = \mathbb{E}_{x \sim \rho} [f(x)] - \beta^{-1} H(\rho). \quad (3.63)$$

through the JKO scheme Jordan et al. (1998):

$$\rho^k = \arg \min_{\rho} \frac{1}{2h} W_2^2(\rho^{k-1}, \rho) + \mathcal{F}(\rho), \quad (3.64)$$

where h is the time step and $W_2(\rho^{k-1}, \rho)$ is the Wasserstein-2 distance between successive distributions. In the non-isotropic case, $D(x) \neq \sigma^2 I$, the induced metric is no longer the Euclidean Wasserstein-2 one, and the potential that is actually minimized deviates from $f(x)$, giving rise to the corrected potential $\Phi(x) \neq f(x)$. It should be noted, as stated in the same paper, that mini-batch matrices typically have a rank lower than 1% of the dimension, which makes them non-isotropic.

If one generalizes to cases of *non-stationarity* also in time, considering thus, with respect to (3.61), $\beta(t)$ (in place of β), one can introduce **Annealing** procedures Kirkpatrick et al. (1983), for which, in the Isotropic case, we know that if $T(t)$, i.e. $\frac{1}{\beta(t)}$, decreases as $\frac{c}{\log(1+t)}$, with c a constant depending on the dimension of the variable under consideration, then we have a convergence condition to the absolute minimum, proven in Hajek (1988). Moreover, given what was stated in (3.61), namely that SGD can be interpreted as a Langevin dynamics, papers such as Sadrtinov et al. (2025) consider $T = T(\eta)$, i.e. a constant temperature depending on the *learning rate*, and moreover an **over-parameterized** network; they seem to state, on the one hand, that this minimizes the Free Energy (3.63), i.e. the one associated with the isotropic case, but they identify, starting from the *train loss*, transitions from chaotic equilibrium to stabilization and then convergence. Since they seem to be considering the isotropic case (even if not made explicit, they have the same functional form F as that case), these transitions seem to point back to those we have analyzed so far Biroli et al. (2024). Indeed, assuming that in a strongly overparameterized regime and with reasonable optimization (3.3) we faithfully reproduce the Score, the missing hypothesis in this analysis would be the thermodynamic limit, but in fact, considering the distribution of weights in this case, I think this is a reasonable assumption.

Note. The results discussed in this section may appear uncorrelated and complex to read together, since they attempt to cover several results from the recent literature on disparate factors related to generative dynamics (of AI in general, we could say). Nevertheless, I preferred to introduce them so that they might serve as a stimulus for further investigations. Finding a thorough theory on the matter would allow us to fully understand the phenomenology associated with generation via Diffusion models and thus, in general, with network-based models, since these too are network-based, like most contemporary AI models, thereby yielding results applicable to more generic cases as well.

3.4 Experimental Validation

In this final part we want to experimentally validate the results obtained from the theoretical analysis of Diffusion Models. In particular we want to present some results, albeit limited to certain cases, regarding the phenomena analyzed in the recent literature on the interpretation of the learning obtained in Diffusion Models through statistical mechanics. In particular we will focus on the phase transitions analyzed in Sections 3.2.2 and 3.2.3. We will then try to provide explanations related to what was introduced in (3.3), namely the results analyzed starting from Han et al. (2026), first investigating the concept of the Over-parameterized regime and the order of magnitude it must actually be, relative to the data, in order to be observed. We then investigated the actual role of the effective dimension of the problem, observing the results on two very different datasets: **MNIST**, a dataset containing 70 thousand images of handwritten digits from 0 to 9, hence with a small d_{eff} , and the **Rademacher** dataset, where we consider images made up solely of equiprobable black and white pixels, and thus very unstructured (large d_{eff}).

3.4.1 Speciation time

Model setup. In this first experiment we consider, as the theoretical model, an isotropic **Gaussian Mixture Model (GMM)** with covariance $\Sigma = \sigma^2 \mathbf{I}$, consisting of $K = 2$ equiprobable clusters centered at $\mathbf{m} = \mathbf{1}_d$ and $-\mathbf{1}_d$. The analysis is repeated for different dimensions:

$$d \in \{256, 1024, 4096, 16384\},$$

in order to study the dependence of the dynamics on dimensionality.

The trajectories, initialized at $T = 10$ (sufficient to approximate the stationary distribution of pure Gaussian noise), are integrated backward down to $t = 0$ by numerically solving the backward SDE:

$$dY_t = \left[Y_t + 2s(Y_t, t) \right] dt + \sqrt{2} d\bar{W}_t,$$

using the **Euler–Maruyama** method with $N = 30$ uniform time steps ($\Delta t = T/N$). The discrete update scheme is:

$$Y_{t_k - \Delta t} = Y_{t_k} (1 + \Delta t) + 2 \Delta t s(Y_{t_k}, t_k) + \sqrt{2 \Delta t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, \mathbf{I}_d),$$

The discretization error $O(\sqrt{\Delta t})$ is small enough not to affect the localization of the critical point t_S . The drift is guided by the **exact score** of the GMM, obtained by analytically differentiating $\log P_t(Y_t)$ (with no approximations due to neural networks):

$$s(Y_t, t) = -\frac{Y_t}{\Delta_t} + \mathbf{m} \frac{e^{-t}}{\Delta_t} \tanh\left(\mathbf{m}^\top Y_t \frac{e^{-t}}{\Delta_t}\right), \quad \Delta_t = 1 - e^{-2t}.$$

Linearization and speciation time. Projecting the dynamics along the direction that separates the two clusters, we define the scalar order variable:

$$y_t = \frac{\mathbf{m}^\top Y_t}{|\mathbf{m}|}. \quad (3.65)$$

For small $|y_t|$ we have $\tanh(z) \approx z$, and the drift linearizes as:

$$\dot{y}_t \approx \lambda(t) y_t, \quad \lambda(t) = 1 - \frac{2}{\Delta_t} + \frac{2|\mathbf{m}|^2 e^{-2t}}{\Delta_t^2}.$$

The sign of $\lambda(t)$ determines the stability of $y = 0$: for $t > t_S$ ($\lambda(t) < 0$) the system remains in the metastable region dominated by noise; for $t < t_S$ ($\lambda(t) > 0$) a pitchfork bifurcation occurs and the trajectory is irreversibly attracted toward $\pm \mathbf{m}$. The critical instant t_S , with $\lambda(t_S) = 0$, identifies the spontaneous symmetry breaking.

Branching and pair dynamics. To identify t_S experimentally, we use *branching*. At different branching times $t_b \in [1, 10]$, the state Y_{t_b} is duplicated into copies A and B , which evolve independently down to $t = 0$ under Brownian noises $\xi_A \neq \xi_B$. The final cluster is determined by the sign of the projection:

$$\text{sign}(\mathbf{m}^\top Y_0).$$

Linearizing the dynamics of the difference $\delta_t = y_t^A - y_t^B$ and modulating the nonlinear term, we obtain:

$$\dot{\delta}_t \approx \left[1 - \frac{2}{\Delta_t} + \frac{2|\mathbf{m}|^2 e^{-2t}}{\Delta_t^2} (1 - \tanh^2(\cdot)) \right] \delta_t + \sqrt{2} (\dot{\xi}_A - \dot{\xi}_B). \quad (3.66)$$

The saturation factor $(1 - \tanh^2(\cdot))$ governs the decoupling: for $t_b > t_S$ the clones choose the minimum independently; for $t_b < t_S$ the difference is damped, and they remain locked to the same pole already selected by the shared trunk.

Order parameter. The order parameter is defined as the probability that two trajectories, cloned at t from the state $\mathbf{y}$, end up in the same class:

$$\phi(t) = \left[\int_{\mathbf{x}_1 \cdot \mathbf{m} > 0, \mathbf{x}_2 \cdot \mathbf{m} > 0} + \int_{\mathbf{x}_1 \cdot \mathbf{m} < 0, \mathbf{x}_2 \cdot \mathbf{m} < 0} \right] d\mathbf{x}_1 d\mathbf{x}_2 d\mathbf{y} P(\mathbf{x}_1, 0 | \mathbf{y}, t) P(\mathbf{x}_2, 0 | \mathbf{y}, t) P(\mathbf{y}, t). \quad (3.67)$$

Numerically it can be estimated by Monte Carlo over concordant pairs:

$$\hat{\phi}(t) = \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} \mathbb{I}(\text{class}(x_{A,i}^{(final)}) == \text{class}(x_{B,i}^{(final)})).$$

In the case of the GMM analyzed, the conditional probability can be rewritten through the forward process:

$$P(\mathbf{x}_1, 0, \mathbf{y}, t) = P_0(\mathbf{x}_1) \frac{e^{-\frac{(\mathbf{y}-\mathbf{x}_1 e^{-t})^2}{2\Delta_t}}}{\sqrt{2\pi\Delta_t}^d}, \quad (3.68)$$

and thus:

$$P(\mathbf{x}_1, 0 \mid \mathbf{y}, t) = P_0(\mathbf{x}_1) \frac{e^{-\frac{(\mathbf{y}-\mathbf{x}_1 e^{-t})^2}{2\Delta_t}}}{\sqrt{2\pi\Delta_t}^d P(\mathbf{y}, t)}. \quad (3.69)$$

Integrating over $\mathbf{x}_1, \mathbf{x}_2$ and factorizing the components orthogonal to $\mathbf{m}$, $\phi(t)$ reduces to the one-dimensional integral in $y = \mathbf{m}^\top \mathbf{Y}_t / |\mathbf{m}|$:

$$\phi(t) = \frac{1}{2} \int dy \frac{G(y, me^{-t}, \Gamma_t)^2 + G(y, -me^{-t}, \Gamma_t)^2}{G(y, me^{-t}, \Gamma_t) + G(y, -me^{-t}, \Gamma_t)}, \quad (3.70)$$

where $G(y, u, v)$ is the scalar Gaussian density with $m = |\mathbf{m}|$ and $\Gamma_t = \Delta_t + \sigma^2 e^{-2t}$.

Asymptotic behavior. For large d , $\phi(t)$ decreases monotonically between the limits:

$$\phi(0) = 1, \quad \phi(t \rightarrow \infty) = \frac{1}{2},$$

reflecting the passage from perfectly correlated states to independent variables. The speciation time t_S is operationally defined by the intermediate threshold:

$$\phi(t_S) = 0.775.$$

Solving this condition for large d recovers the theoretical scaling law (Eq. (3.4.1)), confirming the branching estimate without resorting to simulations:

$$t_S \simeq \frac{1}{2} \log d.$$

Analogy with the Curie–Weiss transition. As anticipated in 3.2.2, the dynamics is analogous to the Curie–Weiss model. The mean force satisfies the self-consistency equation:

$$m_t = \tanh\left(\frac{m_t}{t\sigma^2}\right),$$

while the magnetic susceptibility is:

$$\chi_t = \left. \frac{\partial m_t}{\partial h} \right|_{h=0} = \frac{1 - m_t^2}{t\sigma^2 - (1 - m_t^2)}.$$

The vanishing of the coefficient $\lambda(t)$ identifies t_S exactly as the divergence of χ_t marks the critical temperature of the ferromagnetic transition.

Qualitative Analysis. Introducing the theoretical speciation time:

$$t_S = \alpha \log(d), \quad \alpha = 0.5,$$

Figure 3.4 shows $\phi(t)$ as a function of the normalized time t/t_S . As d varies, all the curves collapse onto a single universal law, crossing the critical region at:

$$\frac{t}{t_S} \approx 1,$$

in agreement with the prediction of Eq. (3.13). As the dimension increases, the transition between the disordered phase ($\phi \simeq 0.5$) and the ordered phase ($\phi \simeq 1$) becomes steeper, taking the form of a discontinuity in the thermodynamic limit $d \rightarrow \infty$.

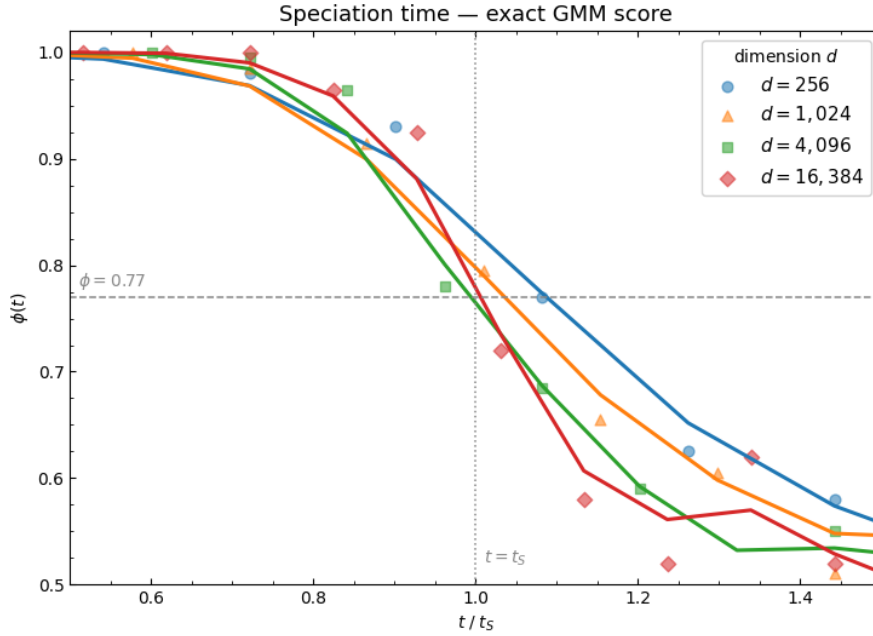


Figure 3.4. Order parameter $\phi(t)$ obtained via branching for different values of the dimension d , using the analytic score of the GMM. Time is normalized with respect to the theoretical value $t_S = \alpha \log d$. The collapse of the curves highlights the universality of the speciation transition, while the equivalent magnetization describes the passage from the disordered phase to the ordered phase.

Verification on individual trajectories. The $\mathbb{Z}_2$ symmetry breaking is also verified on the individual trajectory (Eq. (2.2)) integrated without shared branching, with:

$$d = 512, \quad N_{traj} = 25, \quad T = 8, \quad N_{steps} = 600, \quad \sigma = 1,$$

and the first coordinate $x_1(t)$ aligned with the critical direction. The update scheme is:

$$Y_{t-\Delta t} = Y_t(1 + \Delta t) + 2\Delta t s(Y_t, t) + \sqrt{2\Delta t} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d).$$

For $d = 512$, the theoretical critical limit is $t_S \approx 3.1$. To avoid chromatic artifacts due to high-frequency noise (a consequence of $\lambda(t) \approx 0$ near the critical limit), for $t < t_S$ the color of the branches is assigned by computing a local moving average. Qualitatively, the analysis confirms the stability: for $t > t_S$ the bundles fluctuate freely around the origin with no persistent preference; they separate in the neighborhood of t_S , and then diverge sharply and irreversibly for $t < t_S$.

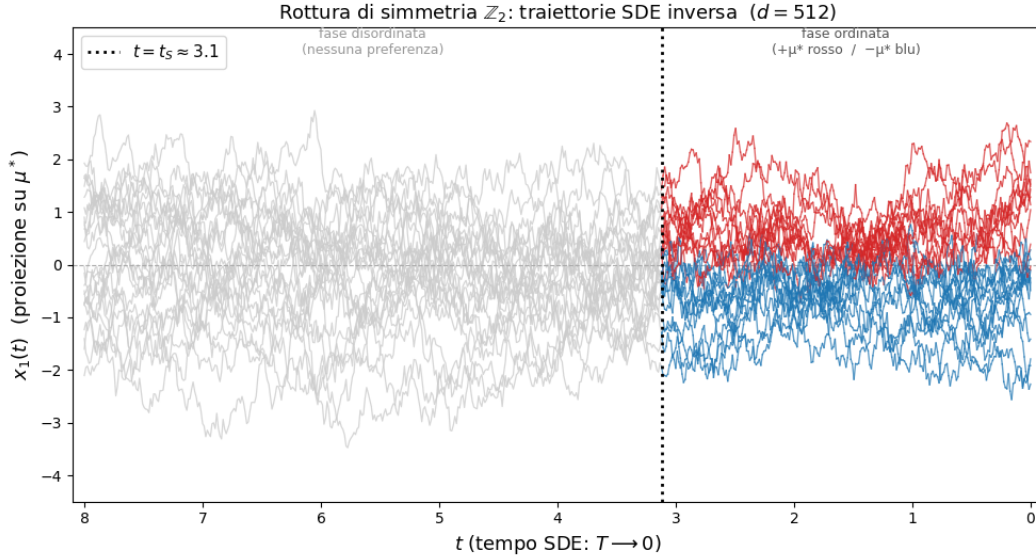


Figure 3.5. Individual trajectories of the reverse SDE in $d = 512$: disordered phase ($t > t_S$, gray) and ordered phase ($t < t_S$, red/blue according to the pole selected). The transition observed trajectory by trajectory is consistent with the theoretical critical instant $t_S \approx 3.1$.

3.4.2 Collapse Time

Considering a dataset generated from a mixture of two Gaussian clusters centered at $\pm \mathbf{m} \in \mathbb{R}^d$ with variance σ^2 , as analyzed in the previous case and with the properties thus described in (3.2.3). As shown analytically, the main marker for identifying the transition to the condensed phase is the excess entropy density $f(t)$, i.e. $f(t) = \psi_{ann}(t) - \psi(t)$, as stated in (3.36), which vanishes in the *annealed* regime (for times below the critical time, i.e. when $t \leq t_c$).

To proceed numerically, $f(t)$ is approximated using a sampling strategy. At a given instant t , n' samples $\mathbf{x}^{(\nu)}(t)$ are drawn by applying the forward diffusion process to the original points of the dataset. The entropy $s(t)$ is then estimated as the empirical average over the generated samples, leading to the following expression for the empirical excess entropy:

$$s^e(t) = -\frac{1}{n'd} \sum_{\nu=1}^{n'} \log P_t^e(\mathbf{x}_t^{(\nu)}), \quad (3.71)$$

where $P_t^e(\mathbf{x})$ represents the empirical probability density associated with the original

dataset:

$$P_t^e(\mathbf{x}) = \frac{1}{n} \sum_{\mu=1}^n \frac{1}{\sqrt{2\pi\Delta_t}^d} \exp\left(-\frac{(\mathbf{x} - \mathbf{a}^\mu e^{-t})^2}{2\Delta_t}\right). \quad (3.72)$$

This approximation is valid and tractable mainly for $t > t_c$. In the experiment we also used:

- $d \in \{256, 1024, 4096\}$.
- $n = 500$ initial training points, split equally into the two clusters centered at $\mathbf{m}$ and $-\mathbf{m}$, with the noise standard deviation set to $\sigma = 0.5$.
- A grid of 60 time steps uniformly distributed in the interval $t \in [0.2, 6.0]$. At each step $n' = 2000$ query samples are drawn, a quantity sufficient to stabilize the average.

At the computational level, direct computation of Equation 3.71 is often unstable for high-dimensional spaces. To address this, the algorithm estimates the normalized fraction $f^e(t)/\alpha$, with $\alpha = \frac{\log n}{d}$, always analyzing the *geometric projections* of the noisy samples onto the signal axis $\mathbf{m}$ (3.65). By subtracting the effect of the analytic noise of variance σ_t^2 from the mean projection, and constraining the result to the interval $[0, 1]$, a clean empirical order curve is obtained. To analytically determine the transition time t_c , the resulting trajectories are finally fitted with a sigmoid function.

Qualitative Results

From the plot we observe that the curves describe the typical dynamics of a phase transition. For short times, the value of $f^e(t)/\alpha$ is approximately zero, confirming that the system is in a disordered symmetric state. Once the threshold time t_c is exceeded, a sudden collapse toward asymptotic saturation at 1 is observed. As d increases (observable moving from the blue curve for $d = 256$ up to the green curve for $d = 4096$), two effects are noted. First, the steepness of the transition curve increases in accordance with the thermodynamic limit ($d \rightarrow \infty$). Second, the empirical critical times (marked by the vertical dashed lines) shift to the right: the larger the spatial dimension, the longer the time (i.e. the amount of noise) required for the target energy to overtake the entropic weight of the entire cluster.

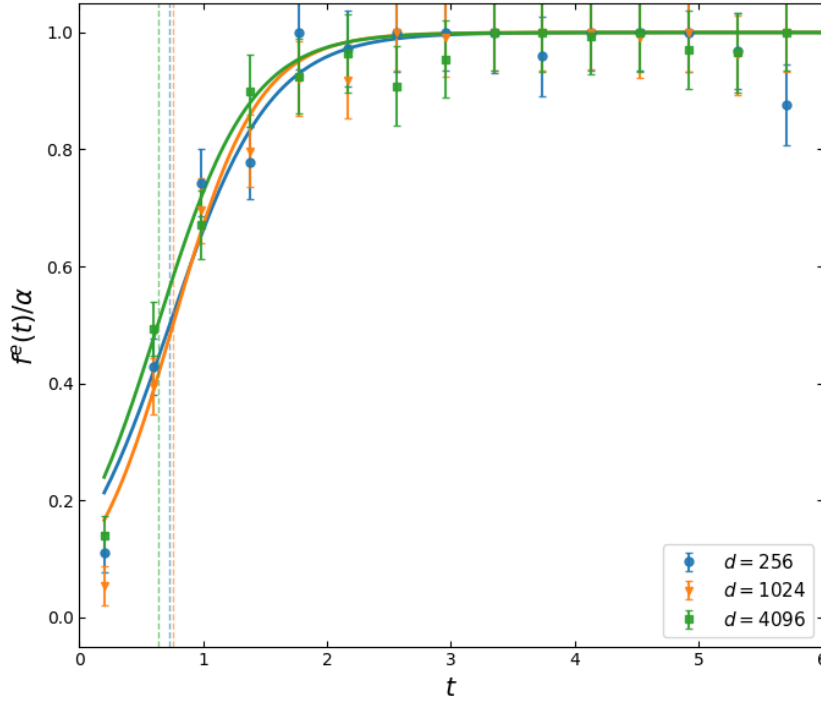


Figure 3.6. Collapse time and Excess Entropy Density. Evolution of the parameter $f^e(t)/\alpha$ as a function of time t for different dimensionalities $d \in \{256, 1024, 4096\}$. Fitting with sigmoid functions highlights the collapse of the curves at the theoretical critical time t_C (indicated by the vertical dashed lines for each d). The figure illustrates the phase transition from the disordered regime (for $t > t_C$, where the reverse SDE is not influenced by the individual samples) to the ordered memorization regime (for $t < t_C$, where the trajectories collapse deterministically onto the training data).

3.4.3 Double Descent

In this experiment we wanted to provide evidence for what was introduced theoretically in (3.3). In particular, we analyzed the behavior of the **Generalization Risk**, defined in (3.5), of which, as stated, we compute a **Monte Carlo (MC)** estimate, namely:

$$\hat{\mathcal{R}} = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} \int_0^T \lambda(t) \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [\|\epsilon - \epsilon_\theta(x_i(t), t)\|^2] dt \quad (3.73)$$

which can be reinterpreted, as done in (3.54), in terms of the bias-variance trade-off:

$$\mathbb{E} [(g_T^{\mathcal{G}}(x) - g^*(x))^2] = \underbrace{(\mathbb{E}[g_T^{\mathcal{G}}(x)] - g^*(x))^2}_{\text{Squared bias}} + \underbrace{\text{Var}(g_T^{\mathcal{G}}(x))}_{\text{Variance}} \quad (3.74)$$

Observing thus the competition between the two terms in a classical interpretation, we could say that as the number of parameters P increases, the variance term increases, since, having a highly parameterized model, in the presence of noisy data, these would be used for their memorization. However, in a more thorough analysis, as done in (3.3), we saw that in fact the optimizers do not allow this. In particular,

we considered the **MNIST** and **Rademacher** datasets, introduced previously, and for both we considered a vector dimension of $d = 784$ (28×28), which coincides with the pixel value of MNIST images. We wanted to analyze the behavior of the **Test Loss** for both cases in order to be able to compare them. Starting from the results obtained, we found the patterns present in the modern literature Lafon and Thomas (2024), which we discussed in (3.3). As we can observe from the plots, near the regime where $P \approx N$ (where P is the number of parameters and $N_{train} = 8.000$ is the number of training data points), we observe a marked rise in the loss function (interpolation peak). However, if the parameter P associated with our network is further increased beyond the interpolation threshold ($P \gg N$), it is possible to observe a second, more marked descent, which brings the loss down to values even lower than those in the first, under-parameterized regime. Moreover, we can see that, provided we use an architecture suited to the manifold of the data Bronstein et al. (2021), if the intrinsic dimension d_{eff} , as in the Rademacher dataset, is large, this second descent is much less pronounced than in a case where the effective dimension is smaller, as in MNIST. This follows what was introduced in (3.56), since we had concluded (without proving it) that the **Generalization Risk** for a Diffusion model scaled on the order of $\mathcal{O}(N^{-\frac{2}{3+d}})$, through the more accurate decomposition (3.57), which also includes the effect due to the optimizer (bias and early stopping). However, the analyses introduced by the authors, we recall, are limited to the case $d = d_{ambient}$ and to the Gradient Descent optimizer. For this analysis we did not resort to numerical methods to try to compute the actual d_{eff} for the two datasets, but relied instead on the hypotheses introduced here for the two cases. In particular, we analyzed how the error of the **Test Loss** scales as the network's capacity varies. The base value p set by us does not directly coincide with the actual number of parameters P , which is computed through polynomial functions dependent on the architecture:

- **Nano U-Net Architecture** for $p \in \{1, 2, 3, 4\}$, where the conversion factor to the actual number of parameters used is:

$$P_{\text{effective}} = 207p^2 + 109p + 4001$$

Carrying out the conversion we obtain:

1. $p = 1 \implies P \approx 4.317, p = 2 \implies P \approx 5.047$
2. $p = 3 \implies P \approx 6.191, p = 4 \implies P \approx 7.749$

- **Standard U-Net Architecture** for $p \in \{1, 2, 3, 4, 6, 8, 12\}$, where the conversion factor to the actual number of parameters varies according to a threshold:

For $p < 4$ (i.e. $p \in \{1, 2, 3\}$):

$$P_{\text{effective}} = 3586p^2 + 502p + 4001$$

From which:

1. $p = 1 \implies P \approx 8.089 \quad (P \approx N_{train})$

$$2. p = 2 \implies P \approx 19.349, p = 3 \implies P \approx 37.781$$

For $p \geq 4$ and EVEN (i.e. $p \in \{4, 6, 8, 12\}$):

$$P_{\text{effective}} = 3590p^2 + 1486p + 1$$

From which:

$$1. p = 4 \implies P \approx 63.385, p = 6 \implies P \approx 138.157$$

$$2. p = 8 \implies P \approx 241.649, p = 12 \implies P \approx 534.793$$

Esp.2 — Double Descent con U-Net DDPM

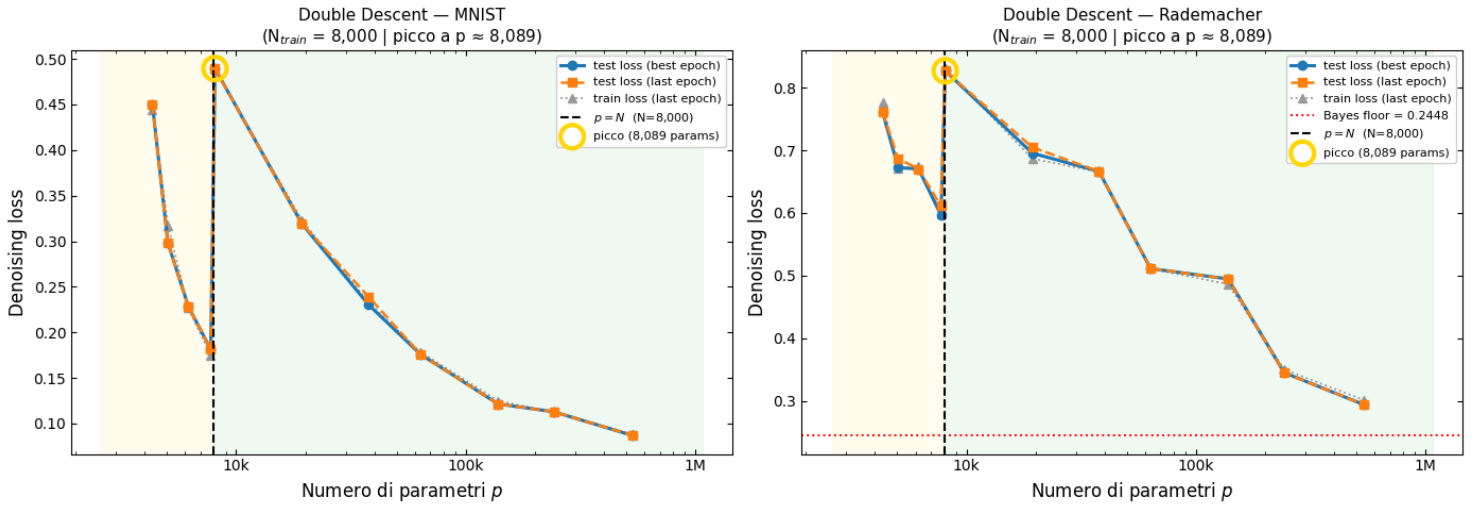


Figure 3.7. Double Descent on MNIST and Rademacher Behavior of the Test Loss as the number of parameters varies, for two problems with different d_{eff} , observing how, in the MNIST case, where d_{eff} is smaller, the Test Loss drops more in the OP regime than in the Rademacher case.

3.4.4 Phase Diagram of P and N

In this final analysis, carried out on the two cases Rademacher and MNIST, we take up again the values of P from the previous case. In particular, for the first experiment we adopted the parameters of the "Nano" model (3.4.3), while for the second we used those of the standard model (3.4.3). In both cases the approach is inspired by the analysis carried out in Bonnaire et al. (2025), where the presence of the three regimes already introduced and analyzed in detail in (3) is distinguished.

Considering specifically N , i.e. the size of the training set ($Train$), for the values $N \in \{1k, 2k, 4k, 6k\}$, we observe how, in the first scenario, the **Test Loss** error decreases as we move further into the **dynamical regime**, and then worsens as we approach the edges of that region. We notice, in particular, rather marked differences between the MNIST case and the Rademacher case; the latter, in fact, remains at high error values even within the generalization regime. This is in line with what is

illustrated in (3.3), since in the first diagram we are analyzing the behavior within the first parametric window. In this region, consistent with what was previously observed for the single case $N = 8000$, the descent of the error as P varies is not particularly pronounced.

However, looking at the second phase diagram in Figure 3.9, we notice how, within this second (overparameterized) window, even the Rademacher case shows an actual descent in terms of the loss function, although less pronounced than in the previous case. Moreover, from the second plot it is interesting to note how, near the **interpolation threshold**, the so-called "second rise of the loss", already introduced in (3.3), emerges. In this scenario, however, it is observed for both datasets that this rise becomes progressively more attenuated as the amount of *Train* data N increases, which could be an interesting subject for further investigation.

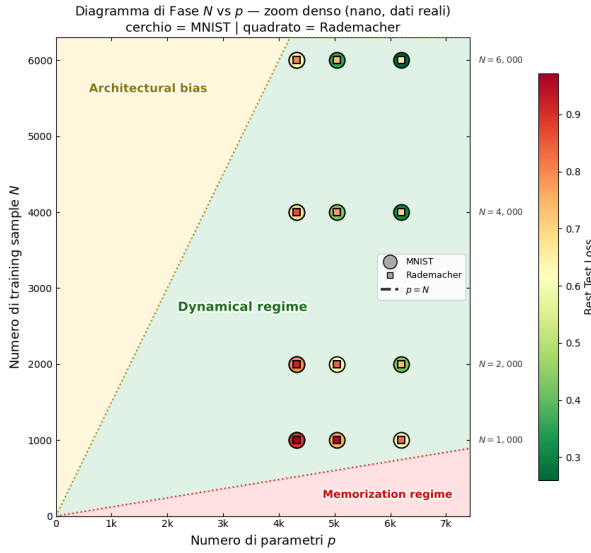
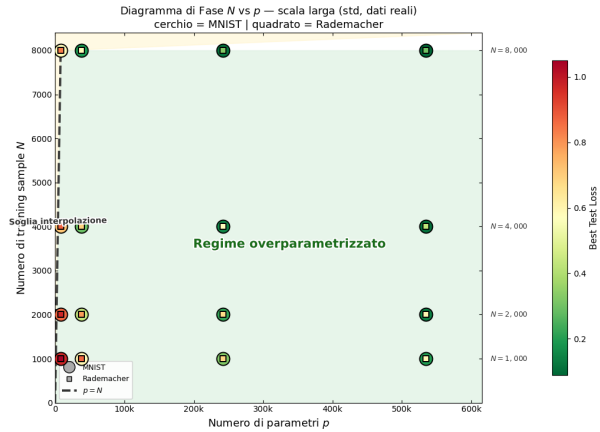


Figure 3.8. Phase diagram of the test loss as (N, P) vary in the first parametric window.

Figure 3.9. Phase diagram of the test loss as (N, P) vary in the second parametric window, i.e. the overparameterized one.



Appendix A

Diffusion Process

Proposition 1 (Derivation of the Fokker-Planck Equation from an SDE).

Consider a one-dimensional stochastic process Y_t governed by the following Itô Stochastic Differential Equation (SDE):

$$dY_t = \mu(Y_t, t)dt + \sigma(Y_t, t)dW_t \quad (\text{A.1})$$

where $\mu(y, t)$ is the drift term, $\sigma(y, t)$ is the volatility (diffusion) term, and W_t is a standard Wiener process. Let $P(y, t)$ be the probability density function of Y_t at time t . Then the time evolution of the density $P(y, t)$ is governed by the partial differential equation known as the Fokker-Planck equation:

$$\frac{\partial P(y, t)}{\partial t} = -\frac{\partial}{\partial y}[\mu(y, t)P(y, t)] + \frac{1}{2}\frac{\partial^2}{\partial y^2}[\sigma^2(y, t)P(y, t)] \quad (\text{A.2})$$

Proof. The proof relies on Itô's Lemma and the use of an arbitrary test function. Let $h(y)$ be a test function that is differentiable at least twice ($h \in C^2(\mathbb{R})$) and compactly supported (it vanishes at infinity together with its derivatives).

Applying Itô's Lemma to the function $h(Y_t)$, we obtain the stochastic differential:

$$dh(Y_t) = h'(Y_t)dY_t + \frac{1}{2}h''(Y_t)(dY_t)^2 \quad (\text{A.3})$$

Substituting dY_t with the expression from the SDE and recalling the rules of Itô calculus ($(dt)^2 = 0$, $dt dW_t = 0$, $(dW_t)^2 = dt$), we obtain:

$$dh(Y_t) = \left[\mu(Y_t, t)h'(Y_t) + \frac{1}{2}\sigma^2(Y_t, t)h''(Y_t) \right] dt + \sigma(Y_t, t)h'(Y_t)dW_t \quad (\text{A.4})$$

Taking the expectation $\mathbb{E}[\cdot]$, the term containing dW_t vanishes since the stochastic integral with respect to a Wiener process is a martingale with zero expectation. We thus obtain an equation for the evolution of the expected value:

$$\frac{d}{dt}\mathbb{E}[h(Y_t)] = \mathbb{E} \left[\mu(Y_t, t)h'(Y_t) + \frac{1}{2}\sigma^2(Y_t, t)h''(Y_t) \right] \quad (\text{A.5})$$

Since the variable Y_t is distributed according to the density $P(y, t)$, we can rewrite the expectations in continuous integral form:

$$\int_{-\infty}^{+\infty} h(y) \frac{\partial P(y, t)}{\partial t} dy = \int_{-\infty}^{+\infty} \left[\mu(y, t)h'(y) + \frac{1}{2}\sigma^2(y, t)h''(y) \right] P(y, t) dy \quad (\text{A.6})$$

At this point, we apply integration by parts to the right-hand side to transfer the spatial derivatives from the test function $h(y)$ to the density function $P(y, t)$. Since $h(y)$ is compactly supported, the boundary terms vanish. For the drift term (a single integration by parts):

$$\int_{-\infty}^{+\infty} h'(y) [\mu(y, t) P(y, t)] dy = - \int_{-\infty}^{+\infty} h(y) \frac{\partial}{\partial y} [\mu(y, t) P(y, t)] dy \quad (\text{A.7})$$

For the diffusion term (two successive integrations by parts):

$$\int_{-\infty}^{+\infty} h''(y) \left[\frac{1}{2} \sigma^2(y, t) P(y, t) \right] dy = \int_{-\infty}^{+\infty} h(y) \frac{1}{2} \frac{\partial^2}{\partial y^2} [\sigma^2(y, t) P(y, t)] dy \quad (\text{A.8})$$

Substituting these expressions into the integral equation and collecting $h(y)$, we obtain:

$$\int_{-\infty}^{+\infty} h(y) \left\{ \frac{\partial P}{\partial t} + \frac{\partial}{\partial y} [\mu P] - \frac{1}{2} \frac{\partial^2}{\partial y^2} [\sigma^2 P] \right\} dy = 0 \quad (\text{A.9})$$

For this integral to vanish for any arbitrary choice of the test function $h(y)$, by the fundamental lemma of the calculus of variations, the expression enclosed in braces must be identically zero almost everywhere. Setting this expression to zero yields exactly the Fokker-Planck equation, which concludes the proof. $\square$

Theorem 1. (Anderson, 1982). Consider a diffusion process of the form (2.11), which we shall from now on call the Forward Process, satisfying the conditions (1.1.2) and the Fokker-Planck equation (1.3). Then the time-reversed process, which we shall from now on refer to as the Backward Process, defined as flowing from $t = T$ to $t = 0$, is also a diffusion process, and is governed by the following *Reverse SDE*:

$$d\bar{X}_t = \left[f(\bar{X}_t, t) - g(t)^2 \nabla_x \log p_t(\bar{X}_t) \right] dt + g(t) d\bar{W}_t \quad (\text{A.10})$$

where the term $\nabla_x \log p_t(x)$ is called the *score function* of the marginal density at time t .

Proof. The proof relies on the macroscopic equivalence dictated by the Fokker-Planck Equation. The time evolution of the marginal density $p_t(x)$ for the *forward* process described by (2.14) is governed by:

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot (f(x, t) p_t(x)) + \frac{1}{2} g(t)^2 \Delta_x p_t(x) \quad (\text{A.11})$$

Using the vector identity $\nabla_x \cdot (p_t(x) \nabla_x \log p_t(x)) = \nabla_x \cdot (\nabla_x p_t(x)) = \Delta_x p_t(x)$, we can rewrite the diffusive term:

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot \left(\left[f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right] p_t(x) \right) \quad (\text{A.12})$$

Let us now consider a generic reverse process described by the SDE $d\bar{X}_t = \tilde{f}(\bar{X}_t, t) dt + g(t) d\bar{W}_t$, where time flows backward. The associated Fokker-Planck Equation, computed with respect to a negative time increment (hence the sign change of the time derivative ∂_t), becomes:

$$-\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot (\tilde{f}(x, t) p_t(x)) + \frac{1}{2} g(t)^2 \Delta_x p_t(x) \quad (\text{A.13})$$

Applying the same vector identity to the Laplacian, we obtain:

$$-\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot \left(\left[\tilde{f}(x, t) - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x) \right] p_t(x) \right) \quad (\text{A.14})$$

In order for the marginal distributions of the forward and backward processes to coincide exactly at every instant t , equations (A.12) and (A.14) must describe the same change in density over time. Equating the effective velocity fields (and taking into account the negative sign of ∂_t), we impose:

$$f(x, t) - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x) = - \left(\tilde{f}(x, t) - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x) \right) \quad (\text{A.15})$$

Solving for the reverse drift $\tilde{f}(x, t)$ we obtain:

$$\tilde{f}(x, t) = f(x, t) - g(t)^2 \nabla_x \log p_t(x) \quad (\text{A.16})$$

Substituting $\tilde{f}$ into the generic reverse SDE yields exactly equation (A.10), which concludes the proof. $\square$

A.1 Forward Process

Proposition 1 (Evolution of the marginal distribution). Let Y_0 be a random variable distributed according to the empirical data distribution $P_0(Y_0) = \frac{1}{N} \sum_{i=1}^N \delta(Y_0 - x_i)$, where $\{x_1, \dots, x_N\}$ is the dataset at our disposal. Let Y_t be the state of the process at time $t > 0$, obtained by perturbing Y_0 with isotropic Gaussian noise such that the transition probability is $P(Y_t|Y_0) = \mathcal{N}(Y_t; Y_0, \sigma_t^2 \mathbf{I})$. Then the marginal distribution $P_t(Y_t)$ takes the form of a Gaussian Mixture Model (GMM):

$$P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \mathcal{N}(Y_t; x_i, \sigma_t^2 \mathbf{I}) \quad (\text{A.17})$$

Proof. To derive the marginal distribution $P_t(Y_t)$ at time t , we apply the law of total probability, marginalizing (i.e., integrating) the joint probability with respect to all possible initial states Y_0 :

$$P_t(Y_t) = \int P(Y_t|Y_0) P_0(Y_0) dY_0$$

Substituting into the integral the known definitions of the transition probability and the initial empirical distribution, we obtain:

$$P_t(Y_t) = \int \mathcal{N}(Y_t; Y_0, \sigma_t^2 \mathbf{I}) \left[\frac{1}{N} \sum_{i=1}^N \delta(Y_0 - x_i) \right] dY_0$$

Exploiting the linearity of the integration operator, we can pull out the summation and the normalization constant:

$$P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \int \mathcal{N}(Y_t; Y_0, \sigma_t^2 \mathbf{I}) \delta(Y_0 - x_i) dY_0$$

We recognize in the integral the convolution operation between a continuous function (the Gaussian) and the Dirac delta. By the sifting property of the Dirac delta, defined as $\int f(x)\delta(x-a)dx = f(a)$, the integral takes the value of the Gaussian function evaluated exactly at the point $Y_0 = x_i$:

$$P_t(Y_t) = \frac{1}{N} \sum_{i=1}^N \mathcal{N}(Y_t; x_i, \sigma_t^2 \mathbf{I})$$

which proves the claim. $\square$

Proposition 2 (Asymptotic convergence). Let Y_t be the Ornstein-Uhlenbeck process (Variance Preserving framework) governed by the stochastic differential equation $dY_t = -Y_t dt + \sqrt{2}dW_t$. Given any initial data distribution $Y_0 \sim P_0(Y_0)$, the marginal distribution of the process Y_t converges in distribution to the standard multivariate normal $\mathcal{N}(\mathbf{0}, \mathbf{I})$ as $t \rightarrow \infty$.

Proof. From equation (2.26), we know that the time evolution of the system, conditioned on having observed the starting point Y_0 , is governed by the Gaussian transition kernel $\mathcal{K}_t(Y_t, Y_0) = \mathcal{N}(Y_t; Y_0 e^{-t}, (1 - e^{-2t})\mathbf{I})$. Recalling that the characteristic function of a generic normal distribution $\mathcal{N}(\mu, \Sigma)$ evaluated at a frequency vector ω is known to be $\exp\left(i\omega^T \mu - \frac{1}{2}\omega^T \Sigma \omega\right)$, we can write the conditional characteristic function of Y_t given Y_0 :

$$\mathbb{E}[e^{i\omega^T Y_t} | Y_0] = \exp\left(i\omega^T (Y_0 e^{-t}) - \frac{1}{2}|\omega|^2(1 - e^{-2t})\right) \quad (\text{A.18})$$

To find the characteristic function of the marginal distribution $P_t(Y_t)$ at an arbitrary time instant t , we apply the law of total expectation, averaging with respect to the (potentially unknown) initial distribution $P_0(Y_0)$:

$$\phi_{Y_t}(\omega) = \mathbb{E}_{Y_t}[e^{i\omega^T Y_t}] = \mathbb{E}_{Y_0}[\mathbb{E}[e^{i\omega^T Y_t} | Y_0]] \quad (\text{A.19})$$

Substituting the expression derived for the action of the kernel:

$$\phi_{Y_t}(\omega) = \mathbb{E}_{Y_0}\left[\exp\left(i\omega^T Y_0 e^{-t} - \frac{1}{2}|\omega|^2(1 - e^{-2t})\right)\right] \quad (\text{A.20})$$

Since the term associated with the stochastic spread (the variance of the kernel) does not depend on the initial state Y_0 , we can factor it out and move it outside the expectation operator:

$$\phi_{Y_t}(\omega) = \exp\left(-\frac{1}{2}|\omega|^2(1 - e^{-2t})\right) \cdot \mathbb{E}_{Y_0}[\exp(i\omega^T Y_0 e^{-t})] \quad (\text{A.21})$$

Let us now analyze the asymptotic behavior of the system by computing the limit as $t \rightarrow \infty$. Knowing that the deterministic contraction drives $e^{-t} \rightarrow 0$ and that the variance saturates at $(1 - e^{-2t}) \rightarrow 1$, we can pass to the limit under the expectation sign. This is formally justified by the Dominated Convergence Theorem (since the complex exponential is trivially bounded in modulus by 1):

$$\lim_{t \rightarrow \infty} \phi_{Y_t}(\omega) = \exp\left(-\frac{1}{2}|\omega|^2\right) \cdot \mathbb{E}_{Y_0}[\exp(0)] = \exp\left(-\frac{1}{2}|\omega|^2\right) \cdot 1 \quad (\text{A.22})$$

The result obtained coincides exactly with the characteristic function of the standard multivariate normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In conclusion, by Lévy's Continuity Theorem, since the marginal characteristic function of Y_t converges pointwise to the characteristic function of $\mathcal{N}(\mathbf{0}, \mathbf{I})$ (which is continuous at the origin), we deduce that the process Y_t converges in distribution to $\mathcal{N}(\mathbf{0}, \mathbf{I})$. This marginalization analytically demonstrates the obliteration of memory: the final result is entirely independent of the topology of the original distribution $P_0(Y_0)$. $\square$

A.2 Backward Process

Proposition 1 (Tweedie's Formula and the optimal estimator). Let $Y_0 \sim q(Y_0)$ be a random variable with an unknown but fixed distribution, and let Y_t be an observation of it corrupted by additive Gaussian noise according to the model:

$$Y_t = Y_0 + \sqrt{\sigma_t^2} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (\text{A.23})$$

equivalently:

$$q(Y_t | Y_0) = \mathcal{N}(Y_t; Y_0, \sigma_t^2 I). \quad (\text{A.24})$$

Let $q(Y_t)$ further denote the marginal distribution:

$$q(Y_t) = \int q(Y_t | Y_0) q(Y_0) dY_0. \quad (\text{A.25})$$

Then the minimum mean square error (MMSE) estimator of the latent variable Y_0 given the observation Y_t is given by:

$$\mathbb{E}[Y_0 | Y_t] = Y_t + \sigma_t^2 \nabla_{Y_t} \log q(Y_t). \quad (\text{A.26})$$

Proof. Consider the Gaussian conditional density:

$$q(Y_t | Y_0) = \mathcal{N}(Y_t; Y_0, \sigma_t^2 I). \quad (\text{A.27})$$

The gradient with respect to Y_t of the logarithm of the density is:

$$\nabla_{Y_t} \log q(Y_t | Y_0) = -\frac{1}{\sigma_t^2} (Y_t - Y_0). \quad (\text{A.28})$$

Multiplying by $q(Y_t | Y_0)$ we obtain:

$$\nabla_{Y_t} q(Y_t | Y_0) = -\frac{1}{\sigma_t^2} (Y_t - Y_0) q(Y_t | Y_0), \quad (\text{A.29})$$

from which it follows that:

$$\sigma_t^2 \nabla_{Y_t} q(Y_t | Y_0) = -(Y_t - Y_0) q(Y_t | Y_0). \quad (\text{A.30})$$

Let us now consider the marginal density:

$$q(Y_t) = \int q(Y_t | Y_0) q(Y_0) dY_0. \quad (\text{A.31})$$

Differentiating under the integral sign:

$$\nabla_{Y_t} q(Y_t) = \int \nabla_{Y_t} q(Y_t | Y_0) q(Y_0) dY_0. \quad (\text{A.32})$$

Multiplying by σ_t^2 :

$$\sigma_t^2 \nabla_{Y_t} q(Y_t) = \int -(Y_t - Y_0) q(Y_t | Y_0) q(Y_0) dY_0. \quad (\text{A.33})$$

Dividing by $q(Y_t)$ and using Bayes' rule:

$$\frac{q(Y_t | Y_0) q(Y_0)}{q(Y_t)} = q(Y_0 | Y_t), \quad (\text{A.34})$$

we obtain:

$$\sigma_t^2 \nabla_{Y_t} \log q(Y_t) = \frac{\sigma_t^2 \nabla_{Y_t} q(Y_t)}{q(Y_t)} \quad (\text{A.35})$$

$$= - \int (Y_t - Y_0) q(Y_0 | Y_t) dY_0. \quad (\text{A.36})$$

Separating the terms:

$$\sigma_t^2 \nabla_{Y_t} \log q(Y_t) = -Y_t \int q(Y_0 | Y_t) dY_0 + \int Y_0 q(Y_0 | Y_t) dY_0. \quad (\text{A.37})$$

Since the first integral equals 1, we obtain:

$$\sigma_t^2 \nabla_{Y_t} \log q(Y_t) = -Y_t + \mathbb{E}[Y_0 | Y_t]. \quad (\text{A.38})$$

Rearranging:

$$\mathbb{E}[Y_0 | Y_t] = Y_t + \sigma_t^2 \nabla_{Y_t} \log q(Y_t). \quad (\text{A.39})$$

□

Proposition 2 (Evidence Lower Bound - ELBO). Let x be an observable datum and z its corresponding latent variable. Given an approximate *encoding* distribution $q_\phi(z|x)$ and a generative joint distribution $p_\theta(x, z) = p_\theta(x|z)p(z)$, the marginal log-likelihood (or evidence) $\log p_\theta(x)$ is rigorously bounded below by a quantity called the **Evidence Lower Bound (ELBO)**. This inequality is expressed by decomposing the ELBO into a reconstruction term and a prior-matching term:

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \mathcal{D}_{\text{KL}}(q_\phi(z|x) \parallel p(z)) \quad (\text{A.40})$$

Proof. Starting from the formal definition of the marginal log-likelihood $\log p_\theta(x)$ and multiplying it by the integral of the approximate distribution $q_\phi(z|x)$ (whose value is 1 over the whole domain), we can expand the function inside the expectation:

$$\log p_\theta(x) = \int q_\phi(z|x) \log p_\theta(x) dz \quad (\text{A.41})$$

$$= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{p_\theta(z|x)} \right] \quad (\text{A.42})$$

Multiplying and dividing the argument of the logarithm by $q_\phi(z|x)$, and exploiting the properties of logarithms to separate the terms, we obtain:

$$\log p_\theta(x) = \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z) q_\phi(z|x)}{p_\theta(z|x) q_\phi(z|x)} \right] \quad (\text{A.43})$$

$$= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)} \right] + \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{q_\phi(z|x)}{p_\theta(z|x)} \right] \quad (\text{A.44})$$

Recognizing in the second term the formal definition of the Kullback-Leibler divergence, we can rewrite the equation as:

$$\log p_\theta(x) = \underbrace{\mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)} \right]}_{\text{ELBO}} + \mathcal{D}_{\text{KL}}(q_\phi(z|x) \parallel p_\theta(z|x)) \quad (\text{A.45})$$

Since the Kullback-Leibler divergence, by Gibbs' inequality, is a strictly non-negative quantity ($\mathcal{D}_{\text{KL}} \geq 0$), we immediately deduce that the ELBO constitutes a lower bound on the evidence.

Finally, applying the product rule to the joint probability $p_\theta(x, z) = p_\theta(x|z)p(z)$ inside the ELBO term, we can isolate the two main functional components of the variational objective:

$$\log p_\theta(x) \geq \text{ELBO} = \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{reconstruction term}} - \underbrace{\mathcal{D}_{\text{KL}}(q_\phi(z|x) \parallel p(z))}_{\text{prior matching term}} \quad (\text{A.46})$$

This inequality completes the proof. $\square$

Proposition 3 (Loss defined from the Evidence Lower Bound). As mentioned previously, the exact computation of the *Score function* (and of the corresponding marginal log-likelihood) turns out to be strongly intractable, since it would require solving an integral over all possible trajectories in the d -dimensional latent space. Historically, this problem of probabilistic intractability had already been encountered in **VAE** models by Kingma and Welling (2013) Kingma and Welling (2013), who overcame the obstacle through the introduction of Variational Inference. Exploiting this framework, we can bypass the exact computation by optimizing a rigorous lower bound, the so-called *Evidence Lower Bound* (ELBO), defined by the following inequality (proved in (2)):

$$\log p_\theta(Y_0) \geq \underbrace{\mathbb{E}_{q_\phi(Y_{1:T}|Y_0)} \left[\log \frac{p_\theta(Y_0, Y_{1:T})}{q_\phi(Y_{1:T}|Y_0)} \right]}_{\text{ELBO}} \quad (\text{A.47})$$

$$= \underbrace{\mathbb{E}_{q_\phi(Y_{1:T}|Y_0)} [\log p_\theta(Y_0|Y_{1:T})]}_{\text{reconstruction term}} - \underbrace{\mathcal{D}_{\text{KL}}(q_\phi(Y_{1:T}|Y_0) \parallel p(Y_{1:T}))}_{\text{prior matching term}} \quad (\text{A.48})$$

Recalling then that our objective is $\max_{\theta} p_\theta(Y_0)$, we observe thanks to the equality (A.45) (from which the $\geq$ in the previous term derives) that:

$$\max_{\theta} \log p_{\theta}(Y_0) = \max_{\theta} \left(\underbrace{\mathbb{E}_{q(Y_{1:T}|Y_0)} \left[\log \frac{p_{\theta}(Y_{0:T})}{q(Y_{1:T}|Y_0)} \right]}_{\text{ELBO}} - \mathcal{D}_{\text{KL}}(q(Y_{1:T}|Y_0) \parallel p_{\theta}(Y_{1:T}|Y_0)) \right). \quad (\text{A.49})$$

Note that, if we assume that the model p_{θ} has sufficient expressive capacity to perfectly approximate the true transitions and that the optimization reaches the global minimum, the entire trajectory of *backward* process transitions will collapse onto that of the *forward* process. At the theoretical optimum, the distance between the two distributions vanishes:

$$\lim_{\text{training} \rightarrow \infty} \mathcal{D}_{\text{KL}}(q(Y_{1:T} \mid Y_0) \parallel p_{\theta}(Y_{1:T} \mid Y_0)) = 0 \quad (\text{A.50})$$

Consequently, maximizing the log-likelihood (LL) is equivalent to maximizing the ELBO. As will be explored further in (3.2.1), for now it suffices to note that in practice models are typically never trained under these assumptions, and therefore quantity (A.49) will never converge to zero; rather, as we will see, it is precisely having $\mathcal{D}_{\text{KL}} > \epsilon$ with $\epsilon > 0$ that allows the model to generalize and avoid collapsing onto the training trajectories, i.e., *overfitting* on the data itself.

Given the above, it is therefore useful to consider the inequality:

$$\log p_{\theta}(Y_0) \geq \mathbb{E}_{q(Y_{1:T}|Y_0)} \left[\log \frac{p(Y_T) \prod_{t=1}^T p_{\theta}(Y_{t-1} \mid Y_t)}{\prod_{t=1}^T q(Y_t \mid Y_{t-1})} \right] \quad (\text{A.51})$$

Expanding the product in the numerator and denominator, and separating the terms for $t = 1$ and $t = T$, we can rewrite the expression as follows:

$$\begin{aligned} \log p_{\theta}(Y_0) &\geq \mathbb{E}_{q(Y_{1:T}|Y_0)} \left[\log \frac{p(Y_T) \prod_{t=1}^T p_{\theta}(Y_{t-1} \mid Y_t)}{\prod_{t=1}^T q(Y_t \mid Y_{t-1})} \right] \\ &= \mathbb{E}_{q(Y_{1:T}|Y_0)} [\log p_{\theta}(Y_0 \mid Y_1)] + \mathbb{E}_{q(Y_{1:T}|Y_0)} \left[\log \frac{p(Y_T)}{q(Y_T \mid Y_{T-1})} \right] \\ &\quad + \sum_{t=1}^{T-1} \mathbb{E}_{q(Y_{1:T}|Y_0)} \left[\log \frac{p_{\theta}(Y_t \mid Y_{t+1})}{q(Y_t \mid Y_{t-1})} \right] \end{aligned}$$

and moreover (proved in Strümke and Langseth (2023)) it holds that:

$$\mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}[f(\mathbf{x}_{a:b})] = \mathbb{E}_{q(\mathbf{x}_{a:b}|\mathbf{x}_0)}[f(\mathbf{x}_{a:b})]$$

and applying this to the first term we obtain:

$$\begin{aligned} &= \mathbb{E}_{q(Y_1|Y_0)} [\log p_{\theta}(Y_0 \mid Y_1)] + \mathbb{E}_{q(Y_{T-1:T}|Y_0)} \left[\log \frac{p(Y_T)}{q(Y_T \mid Y_{T-\Delta t})} \right] \\ &\quad + \sum_{t=1}^{T-1} \mathbb{E}_{q(Y_{t-1:t+1}|Y_0)} \left[\log \frac{p_{\theta}(Y_t \mid Y_{t+1})}{q(Y_t \mid Y_{t-1})} \right] \quad (\text{A.52}) \end{aligned}$$

Recalling the definition of the KL divergence, $\mathcal{D}_{KL}(q \parallel p) = \mathbb{E}_q \left[\log \frac{q}{p} \right]$, we can flip the fractions inside the logarithms. This procedure introduces negative signs, allowing us to simplify the previous equation:

$$\begin{aligned} \log p_\theta(Y_0) &\geq \underbrace{\mathbb{E}_{q(Y_1|Y_0)} [\log p_\theta(Y_0|Y_1)]}_{\text{Reconstruction}} - \underbrace{\mathbb{E}_{q(Y_{T-1}|Y_0)} [\mathcal{D}_{KL}(q(Y_T|Y_{T-1}) \parallel p_\theta(Y_T))]}_{\text{Prior matching}} \\ &\quad - \sum_{t=1}^{T-1} \underbrace{\mathbb{E}_{(Y_{t-1}, t+1|Y_0)} [\mathcal{D}_{KL}(q(Y_t|Y_{t-1}) \parallel p_\theta(Y_t|Y_{t+1}))]}_{\text{Consistency terms}} \end{aligned} \quad (\text{A.53})$$

Notice, however, that in the *Consistency terms* the term (A.55) appears, which we said was intractable. Nevertheless, in Sohl-Dickstein et al. (2015) a fundamental probabilistic device is introduced, which we mentioned earlier. Indeed, in the diffusion model we can describe the process through a Markov chain, as stated in (1.1.2), and knowing that $p(Y_T) = q(Y_T) = \mathcal{N}(Y_T; \mathbf{0}, \mathbf{I})$, we can rewrite the joint probability of the global trajectory in the opposite direction using Bayes' rule:

$$q(Y_{1:T}|Y_0) = q(Y_T|Y_0) \prod_{t=2}^T q(Y_{t-1}|Y_t, Y_0) \quad (\text{A.54})$$

Although the generative backward process must reconstruct a data point starting from noise, from the *forward* trajectories the initial datum $\mathbf{Y}_0$ is perfectly known. By also conditioning the reverse transition probability on this quantity, we can apply Bayes' rule to define the so-called *forward posterior*:

$$q(Y_{t-\Delta t} | Y_t, Y_0) = \frac{q(Y_{t-\Delta t} | Y_0)q(Y_t | Y_{t-\Delta t}, Y_0)}{q(Y_t | Y_0)} = \frac{q(Y_{t-\Delta t} | Y_0)q(Y_t | Y_{t-\Delta t})}{q(Y_t | Y_0)} \quad (\text{A.55})$$

We can now note that the three terms of the fraction are all known distributions:

1. $q(Y_{t-\Delta t} | Y_0) = \mathcal{N}(Y_{t-\Delta t}; Y_0 e^{-(t-\Delta t)}, (1 - e^{-2(t-\Delta t)})\mathbf{I})$
2. $q(Y_t | Y_{t-\Delta t}) = \mathcal{N}(Y_t; Y_{t-\Delta t} e^{-\Delta t}, (1 - e^{-2\Delta t})\mathbf{I})$
3. $q(Y_t | Y_0) = \mathcal{N}(Y_t; Y_0 e^{-t}, (1 - e^{-2t})\mathbf{I})$

Moreover, since the product and the ratio of Gaussians are themselves Gaussian, the intractable ideal reverse process turns into an exact, analytically computable mathematical target: a Normal distribution $q(Y_{t-\Delta t} | Y_t, Y_0) \sim \mathcal{N}(Y_{t-1}; \tilde{\mu}_t(Y_t, Y_0), \tilde{\beta}_t \mathbf{I})$, with $\tilde{\beta}_t = \frac{(1-e^{-2\Delta t})(1-e^{-2(t-\Delta t)})}{1-e^{-2t}}$, which the network will have to approximate. In this formulation, we therefore focus on predicting the mean while fixing the covariance of the distribution, further assuming for the matrix a form implying data independence (its estimation, however, often leads to qualitatively better results in generation Nichol and Dhariwal (2021)). Applying the decomposition to the *prior matching* term, knowing that the target distribution is a

standard normal $p_\theta(Y_T) = p(Y_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and that the argument of the divergence no longer depends on Y_{T-1} , we can rewrite the term as:

$$\begin{aligned}
& \mathbb{E}_{q(Y_{T-1}|Y_0)} [\mathcal{D}_{KL}(q(Y_T|Y_{T-1}) \parallel p_\theta(Y_T))] \\
&= \mathbb{E}_{q(Y_{T-1}|Y_0)} [\mathcal{D}_{KL}(q(Y_T|Y_0) \parallel p(Y_T))] \\
&= \mathcal{D}_{KL}(q(Y_T|Y_0) \parallel p(Y_T)) \underbrace{\int q(Y_{T-1}|Y_0) dY_{T-1}}_{=1} \\
&= \mathcal{D}_{KL}(q(Y_T|Y_0) \parallel p(Y_T))
\end{aligned} \tag{A.56}$$

Realigining the summation window from $[1, T-1]$ to $[2, T]$, for the *consistency terms*, we obtain from (A.53) its tractable form:

$$\begin{aligned}
\log p_\theta(Y_0) &\geq \underbrace{\mathbb{E}_{q(Y_1|Y_0)} [\log p_\theta(Y_0|Y_1)]}_{\text{Reconstruction term}} \\
&\quad - \underbrace{\mathcal{D}_{KL}(q(Y_T|Y_0) \parallel p(Y_T))}_{\text{Prior matching term}} \\
&\quad - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(Y_t|Y_0)} [\mathcal{D}_{KL}(q(Y_{t-1}|Y_t, Y_0) \parallel p_\theta(Y_{t-1}|Y_t))]}_{\text{Denoising matching terms}}
\end{aligned} \tag{A.57}$$

Observing that the *Prior matching term* does not depend on the parameter θ , our optimization problem reduces to:

$$\max_{\theta} \left(\underbrace{\mathbb{E}_{q(Y_1|Y_0)} [\log p_\theta(Y_0 | Y_1)]}_{\text{Reconstruction term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(Y_t|Y_0)} [\mathcal{D}_{KL}(q(Y_{t-1} | Y_t, Y_0) \parallel p_\theta(Y_{t-1} | Y_t))]}_{\text{Denoising matching terms}} \right) \tag{A.58}$$

Setting aside for the moment only the *Reconstruction term*, which we will analyze more carefully in (A.2), we obtain:

$$\begin{aligned}
&= \min_{\theta} \left(\sum_{t=2}^T \mathbb{E}_{q(Y_t|Y_0)} [\mathcal{D}_{KL}(q(Y_{t-1}|Y_t, Y_0) \parallel p_\theta(Y_{t-1}|Y_t))] \right) \\
&= \min_{\theta} (\mathbb{E}_{q(Y_t|Y_0)} [\mathcal{D}_{KL}(q(Y_{t-1}|Y_t, Y_0) \parallel p_\theta(Y_{t-1}|Y_t))])
\end{aligned} \tag{A.59}$$

which we can thus rewrite as:

$$\mathcal{D}_{KL}(q(Y_{t-1}|Y_t, Y_0) \parallel p_\theta(Y_{t-1}|Y_t)) = \int q(Y_{t-1}|Y_t, Y_0) \log \frac{q(Y_{t-1}|Y_t, Y_0)}{p_\theta(Y_{t-1}|Y_t)} dY_{t-1} \tag{A.60}$$

Now expanding the definition of the KL divergence, we introduce a second integral over the variable Y_{t-1} :

$$= \min_{\theta} \int q(Y_t|Y_0) \left[\int q(Y_{t-1}|Y_t, Y_0) \log \frac{q(Y_{t-1}|Y_t, Y_0)}{p_\theta(Y_{t-1}|Y_t)} dY_{t-1} \right] dY_t. \tag{A.61}$$

At this point we exploit the Gaussian parametrization of both the *forward posterior* and the reverse model, where by assumption we have fixed the covariance to $\tilde{\beta}_t \mathbf{I}$ (equivalent to assuming independent data):

$$= \min_{\theta} \int q(Y_t|Y_0) \left[\int \mathcal{N}(Y_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t \mathbf{I}) \log \frac{\mathcal{N}(Y_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t \mathbf{I})}{\mathcal{N}(Y_{t-1}; \mu_{\theta}, \tilde{\beta}_t \mathbf{I})} dY_{t-1} \right] dY_t. \quad (\text{A.62})$$

This amounts to computing the KL divergence between two multivariate Normal distributions:

$$= \min_{\theta} \int q(Y_t|Y_0) \mathcal{D}_{KL}(\mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t \mathbf{I}) \parallel \mathcal{N}(\mu_{\theta}, \tilde{\beta}_t \mathbf{I})) dY_t. \quad (\text{A.63})$$

Applying now the closed-form expression for the KL divergence between multivariate Gaussians, we obtain:

$$= \min_{\theta} \int q(Y_t|Y_0) \left[\frac{1}{2} \left(\text{tr}(\Sigma^{-1} \Sigma) + (\mu_{\theta} - \tilde{\mu}_t)^T \Sigma^{-1} (\mu_{\theta} - \tilde{\mu}_t) - d + \log \frac{|\Sigma|}{|\tilde{\Sigma}|} \right) \right] dY_t, \quad (\text{A.64})$$

Thanks to this choice, the trace and log-determinant terms simplify drastically:

$$\text{tr}(\Sigma^{-1} \Sigma) = d, \quad \log \frac{|\Sigma|}{|\tilde{\Sigma}|} = 0. \quad (\text{A.65})$$

Consequently, the expression reduces to:

$$= \min_{\theta} \int q(Y_t|Y_0) \left[\frac{1}{2} \left(d + \frac{1}{\tilde{\beta}_t} \|\mu_{\theta} - \tilde{\mu}_t\|_2^2 - d \right) \right] dY_t. \quad (\text{A.66})$$

Cancelling the constant terms d , which do not depend on θ , we finally obtain:

$$= \min_{\theta} \int q(Y_t|Y_0) \left[\frac{1}{2\tilde{\beta}_t} \|\mu_{\theta}(Y_t, t) - \tilde{\mu}_t(Y_t, Y_0)\|_2^2 \right] dY_t. \quad (\text{A.67})$$

and returning to the compact expectation notation:

$$= \min_{\theta} \mathbb{E}_{q(Y_t|Y_0)} \left[\frac{1}{2\tilde{\beta}_t} \|\mu_{\theta}(Y_t, t) - \tilde{\mu}_t(Y_t, Y_0)\|_2^2 \right]. \quad (\text{A.68})$$

Thus obtaining a weighted *Mean Squared Error* of the means, and recalling (A.55), we can write the mean and variance of $q(Y_{t-1}|Y_t, Y_0) = \mathcal{N}(Y_{t-1}; \tilde{\mu}_t(Y_t, Y_0), \tilde{\beta}_t \mathbf{I})$ as:

$$\tilde{\mu}_t(Y_t, Y_0) = \frac{e^{-(t-\Delta t)}(1 - e^{-2\Delta t})}{1 - e^{-2t}} Y_0 + \frac{e^{-\Delta t}(1 - e^{-2(t-\Delta t)})}{1 - e^{-2t}} Y_t \quad \tilde{\beta}_t = \frac{1 - e^{-2(t-\Delta t)}}{1 - e^{-2t}} (1 - e^{-2\Delta t}) \quad (\text{A.69})$$

1

However, during the generative reverse process the original datum Y_0 is not known. The solution consists of training a neural network, parametrized by θ , to produce an estimate $\hat{Y}_\theta(Y_t, t)$ of the original datum starting from the noisy observation Y_t .

We can therefore parametrize the mean of our model $\mu_\theta(Y_t, t)$ by mimicking the structure of the exact equation, but replacing the unknown Y_0 with its approximation:

$$\mu_\theta(Y_t, t) = \frac{e^{-(t-\Delta t)}(1 - e^{-2\Delta t})}{1 - e^{-2t}} \hat{Y}_\theta(Y_t, t) + \frac{e^{-\Delta t}(1 - e^{-2(t-\Delta t)})}{1 - e^{-2t}} Y_t \quad (\text{A.70})$$

Substituting the expressions for $\tilde{\mu}_t$ and μ_θ into the objective defined in (A.68), we notice that the terms depending on Y_t cancel out perfectly. Collecting the constants, the *loss* simplifies into a direct quadratic difference between the true datum and the network's prediction: Substituting the exact definitions of the means $\tilde{\mu}_t$ and μ_θ into the Kullback-Leibler Divergence, we notice that the terms depending on the current state Y_t are identical and cancel each other out. The divergence thus reduces to the squared Euclidean distance between the true datum and the network's prediction, scaled by the variance and by the square of the proportionality coefficient:

$$= \mathbb{E}_{q(Y_t|Y_0)} \left[\frac{1}{2\tilde{\beta}_t} \left(\frac{e^{-(t-\Delta t)}(1 - e^{-2\Delta t})}{1 - e^{-2t}} \right)^2 \|Y_0 - \hat{Y}_\theta(Y_t, t)\|_2^2 \right] \quad (\text{A.71})$$

Instead of having the model predict the original image Y_0 , as observed in Denoising Diffusion Probabilistic Models (DDPM) and Denoising Diffusion Implicit Models (DDIMs), Ho et al. (2020); Nichol and Dhariwal (2021), it turns out to be more effective to reformulate the objective by isolating Y_0 from equation (A.70) (with a change of notation) as:

$$Y_0 = e^t \left(Y_t - \sqrt{1 - e^{-2t}} \epsilon \right) \quad (\text{A.72})$$

Substituting this expression into the definition of the exact mean $\tilde{\mu}_t$, we obtain a formulation depending only on the current state Y_t and the noise ϵ_t :

$$\tilde{\mu}_t(Y_t, \epsilon) = e^{\Delta t} \left(Y_t - \frac{1 - e^{-2\Delta t}}{\sqrt{1 - e^{-2t}}} \epsilon \right) \quad (\text{A.73})$$

Consequently, it is possible to reparametrize the model μ_θ so that the neural network directly estimates the added noise, denoted as $\hat{\epsilon}_\theta(Y_t, t)$:

$$\mu_\theta(Y_t, t) = e^{\Delta t} \left(Y_t - \frac{1 - e^{-2\Delta t}}{\sqrt{1 - e^{-2t}}} \epsilon_\theta(Y_t, t) \right) \quad (\text{A.74})$$

Computing the squared difference between these two means in the *loss*, the terms in Y_t cancel out. The objective thus becomes the minimization of the noise estimation error:

$$\mathcal{L}_{t-1} = \mathbb{E}_{q(Y_t|Y_0)} \left[\frac{(1 - e^{-2\Delta t})^2}{2\tilde{\beta}_t e^{-2\Delta t} (1 - e^{-2t})} \|\epsilon - \epsilon_\theta(Y_t, t)\|_2^2 \right] \quad (\text{A.75})$$

¹For brevity, the complete algebraic steps for the derivation of $\tilde{\mu}_t$ and $\tilde{\beta}_t$ are omitted. Since these are standard results based on the properties of Gaussian distributions, the reader is referred to Ho et al. (2020) for the extended proof. Furthermore, in some passages the term $t - \Delta t$ is denoted as $t - 1$, purely to make the steps more explanatory through the theory of Markov chains.

Finally, as demonstrated empirically by Ho et al. (2020), removing the constant weighting factor improves training stability and quality, thus defining the final *Simplified Loss*:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{q(Y_t|Y_0)} \left[\|\hat{\epsilon}_\theta(Y_t, t) - \epsilon_t\|_2^2 \right] \quad (\text{A.76})$$

Analysis of the *Reconstruction Term*. Were the *Reconstruction term* not omitted, the complete objective function would include an additional contribution evaluated at the final instant $t = 1$, defined as:

$$\mathcal{L}_{\text{rec}} = -\mathbb{E}_{q(Y_1|Y_0)} [\log p_\theta(Y_0 | Y_1)]. \quad (\text{A.77})$$

Under the assumption of a Gaussian parametrization of the type $p_\theta(Y_0 | Y_1) = \mathcal{N}(Y_0; \mu_\theta(Y_1, 1), \sigma^2 \mathbf{I})$, this term reduces (up to additive constants independent of the parameters θ) to a quadratic loss:

$$\mathcal{L}_{\text{rec}} \propto \mathbb{E}_{q(Y_1|Y_0)} [\|Y_0 - \mu_\theta(Y_1, 1)\|_2^2]. \quad (\text{A.78})$$

This expression indeed corresponds to a Mean Squared Error computed on the direct reconstruction of the original datum Y_0 starting from the noisy state Y_1 .

However, since this contribution acts exclusively on the last time step ($t = 1$), while the remaining terms of the loss function constrain the reverse dynamics along the entire trajectory for $t \geq 2$, the reconstruction term turns out to be analytically redundant with respect to the *denoising matching terms*, which already implicitly guarantee the correct approximation of the marginal data distribution.

Consequently, in standard practice, $\mathcal{L}_{\text{rec}}$ tends to be neglected during optimization, focusing training exclusively on the *denoising* terms distributed over the entire time horizon.

Proposition 4 (Equivalence between the Loss defined from the ELBO and the Denoising Score Matching Loss). Consider the Loss obtained through extremization of the ELBO function, which, as established in (3), coincides with:

$$\mathcal{L}_{\text{ELBO}} \propto \|\mu_\theta(Y_t, t) - \tilde{\mu}_t(Y_t, Y_0)\|^2 \quad (\text{A.79})$$

However, the term $\tilde{\mu}_t$, recalling how we defined $P_t(Y_t|Y_0)$, can be rewritten as:

$$\tilde{\mu}_t = e^{\Delta t} Y_t - e^{\Delta t} \frac{1 - e^{-2\Delta t}}{\sqrt{1 - e^{-2t}}} \epsilon \quad (\text{A.80})$$

and substituting what was obtained in (2.2) as the estimate given by Tweedie's formula, namely $\epsilon = -\sqrt{1 - e^{-2t}} \nabla_{Y_t} \log q(Y_t|Y_0)$, we obtain that:

$$\tilde{\mu}_t = e^{\Delta t} Y_t + e^{\Delta t} (1 - e^{-2\Delta t}) \nabla_{Y_t} \log q(Y_t|Y_0) \quad (\text{A.81})$$

and since we can furthermore write the term μ_θ as:

$$\mu_\theta = e^{\Delta t} Y_t + e^{\Delta t} (1 - e^{-2\Delta t}) s_\theta(Y_t, t) \quad (\text{A.82})$$

obtaining then from (A.80):

$$\mathcal{L} \propto \left\| \left[e^{\Delta t} Y_t + e^{\Delta t} (1 - e^{-2\Delta t}) s_\theta \right] - \left[e^{\Delta t} Y_t + e^{\Delta t} (1 - e^{-2\Delta t}) \nabla \log q \right] \right\|^2 \quad (\text{A.83})$$

which therefore corresponds to:

$$\mathcal{L} \propto \lambda(t) \|s_\theta(Y_t, t) - \nabla_{Y_t} \log q(Y_t|Y_0)\|^2 \quad (\text{A.84})$$

from which we derive the equivalence.

Appendix B

Learning the Generative Process

Proposition 1. Curse of Dimensionality In (1.1.4) we mentioned the computational problem associated with the grid required for a high-dimensional PDE, introduced by Richard Bellman Bellman (1961). We now wish to address the problem from a statistical and geometric point of view related to the distribution of data in space.

Proof. Let d be the number of *features* associated with a data point. Since we normalize the values taken by these variables, we can imagine the data as contained within the hypercube $[-1, 1]^d$. The volume of this hypercube in this case is

$$V_{cube} = 2^d.$$

To better understand the geometric distribution of data in high-dimensional space, it is useful to compare this volume with that of a d -dimensional hypersphere defining the distance between our data points.

Let us then consider the d -dimensional ball of radius R

$$B_d(R) = \{x \in \mathbb{R}^d : \|x\| \leq R\}.$$

Its volume is defined as

$$V_d(R) = \int_{B_d(R)} 1 \, dx.$$

To compute this integral, it is convenient to switch to polar coordinates in d dimensions. $dx = r^{d-1} dr \, d\Omega$.

The volume then becomes

$$V_d(R) = \int_{S^{d-1}} d\Omega \int_0^R r^{d-1} dr.$$

hence

$$V_d(R) = \frac{R^d}{d} \int_{S^{d-1}} d\Omega.$$

from which we obtain the general formula for the volume of the d -dimensional ball, which in the particular case of unit radius ($R = 1$) is defined as:

$$V_d = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2} + 1\right)}.$$

If we then consider:

$$\frac{V_{sphere}}{V_{cube}} = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2} + 1\right)} \frac{1}{2^d}.$$

Applying Stirling's approximation for $\Gamma\left(\frac{d}{2} + 1\right) \sim \sqrt{\pi d} \left(\frac{d}{2e}\right)^{d/2}$ we obtain:

$$\frac{V_{sphere}}{V_{cube}} \sim \frac{\pi^{d/2}}{2^d \sqrt{\pi d} \left(\frac{d}{2e}\right)^{d/2}} = \frac{1}{\sqrt{\pi d}} \left(\frac{2\pi e}{2d}\right)^{d/2} = \frac{1}{\sqrt{\pi d}} \left(\frac{\pi e}{d}\right)^{d/2}.$$

This shows that for large d the fraction of space occupied by the sphere rapidly tends to zero.

To represent the distribution in space with resolution $\varepsilon > 0$, we divide each dimension into intervals of side ε . Each cell has volume ε^d , and the total number of cells needed to cover the cube is:

$$N \sim \frac{V_{cube}}{\varepsilon^d} = \frac{2^d}{\varepsilon^d} = \left(\frac{2}{\varepsilon}\right)^d.$$

Defining $c = 2/\varepsilon > 1$, we obtain the exponential growth typical of the *curse of dimensionality*:

$$N = O(c^d).$$

From a statistical point of view, this phenomenon has a fundamental consequence: to adequately describe a distribution in $\mathbb{R}^d$, a number of samples growing exponentially with d is required. An important consequence is that real data therefore rarely occupy the whole space $\mathbb{R}^d$ uniformly, but tend to concentrate on geometric structures of much lower intrinsic dimension, often modeled as manifolds embedded in the high-dimensional space. This observation constitutes one of the theoretical motivations underlying modern generative models, including diffusion models. $\square$

Proposition 2. Analysis of the empirical data matrix through RMT.

Consider $D^t \in \mathbb{R}^{d \times d}$ such that we can decompose each element of the matrix D^t as its exact expectation value, namely C^t , plus a statistical fluctuation:

$$D_{i,j}^t = C_{i,j}^t + \delta C_{i,j}^t. \quad (\text{B.1})$$

then, by the **Central Limit Theorem**, we know that the error scales as $\frac{1}{\sqrt{N}}$, and therefore:

$$\delta C^t = \frac{1}{\sqrt{N}} W, \quad (\text{B.2})$$

with $W \in \mathbb{R}^{d \times d}$ a random matrix.

A fundamental result in Random Matrix Theory (RMT) concerns sample covariance matrices, whose spectrum is asymptotically governed by the *Marchenko-Pastur law*. However, to study the limit of the error fluctuations alone, it is customary to

resort to an approximation based on *Wigner's semicircle law* Anderson et al. (2010), which states that under the assumptions that:

1. The elements W_{ij} for $i \leq j$ are i.i.d. random variables.
2. The off-diagonal elements have mean 0 and finite variance.
3. The diagonal elements have finite variance.

the largest eigenvalue scales as $\max_{\text{spec}(W)} = O(\sqrt{d})$.

In our case, the fluctuation matrix is:

$$\delta C_{ij}^t = \frac{1}{N} \sum_{\mu=1}^N \left(x_i^\mu(t) x_j^\mu(t) - \mathbb{E}[x_i(t) x_j(t)] \right), \quad (\text{B.3})$$

and thus, by the **CLT**,

$$\sqrt{N} \delta C_{i,j}^t \sim \mathcal{N}(0, \sigma_{i,j}^2), \quad (\text{B.4})$$

which guarantees zero mean and finite variance, satisfying assumptions 2 and 3.

As for the first assumption, strictly speaking the components on the same row or column share variables (falling within the Wishart ensemble) and are not independent for covariance matrices. However, in high dimension $d \rightarrow \infty$, two random vectors tend to become almost orthogonal:

$$\text{Var} \left[\sum_{i=1}^d u_i v_i \right] = O(d) \quad \Rightarrow \quad \frac{1}{d} \mathbf{u} \cdot \mathbf{v} \rightarrow 0 \text{ by the law of large numbers.} \quad (\text{B.5})$$

This asymptotic orthogonality induces a strong decorrelation between entries. Consequently, in the thermodynamic limit, we can assume an equivalent Gaussian model and consider, in the limit, the first assumption to also be satisfied for the error matrix.

We then define, in order to exploit the eigenvalue scaling, a normalized Wigner matrix:

$$R^t = \frac{1}{\sqrt{d}} W, \quad (\text{B.6})$$

and note that:

$$\max_{\text{spec}(R^t)} = \frac{1}{\sqrt{d}} \cdot \mathcal{O}(\sqrt{d}) = \mathcal{O}(1). \quad (\text{B.7})$$

Substituting into the equation for δC^t and approximating the error term as:

$$\delta C^t \approx \sqrt{\frac{d}{N}} R^t. \quad (\text{B.8})$$

We thus obtain the perturbative model for the empirical matrix:

$$D^t = C^t + \sqrt{\frac{d}{N}} R^t. \quad (\text{B.9})$$

The elements of R^t have zero mean and variance $\mathcal{O}(1/d)$, and, consistently with what was stated in (B.7), their random geometry ensures that no spatial direction is

privileged. Consequently, if we consider the constraint imposed by the unit Euclidean norm of the eigenvectors ($\|\mathbf{u}\|_2^2 = \sum_{i=1}^d u_i^2 = 1$), we find that the energy is distributed homogeneously across the d coordinates. This forces the individual components to scale asymptotically as $\mathcal{O}(1/\sqrt{d})$, which implies that the eigenvectors are *delocalized* by this factor with respect to the exact basis of C^t .

If we now consider (B.9), and in particular rewrite $N = d^{a+1}$, we obtain

$$D^t = C^t + \frac{1}{d^{a/2}} R^t = C^t + \mathcal{O}(d^{-a/2}) R^t, \quad (\text{B.10})$$

Proposition 3. Learning the eigenvalue density for $a > 0$. For a rigorous understanding of this result the reader is referred to Biroli and Mezard (2023); however, an intuitive idea can be obtained by considering the following generating function:

$$G(z) = \frac{1}{d} \text{Tr} \left(\frac{1}{zI - C^t} \right), \quad (\text{B.11})$$

which we interpret as the moment generating function of the spectral distribution. Its mass, i.e., the macroscopic probability density of the eigenvalues $\rho(\lambda)$, can be obtained by evaluating the imaginary part of the function near the real axis:

$$\rho(\lambda) = \frac{1}{\pi} \lim_{\epsilon \rightarrow 0^+} \text{Im} G(\lambda - i\epsilon). \quad (\text{B.12})$$

This relation is known in the literature as the Stieltjes inversion formula (or the Perron-Frobenius-Sokhotski theorem). Substituting the perturbative equation (B.10) (i.e., $D^t = C^t + d^{-a/2} R^t$) into the original definition of the resolvent, we obtain:

$$G_D(z) = \frac{1}{d} \text{Tr} \left(\frac{1}{zI - (C^t + d^{-a/2} R^t)} \right). \quad (\text{B.13})$$

One observes that, if this operator is formally expanded, the random noise R^t introduces correction terms depending on the projections onto its own eigenvectors. Since, as discussed previously, the eigenvectors are delocalized, then under the trace operator (which performs a spatial average), these incoherent fluctuations cancel out asymptotically, bounding (B.13) from above. Consequently, in the limit $d \rightarrow \infty$, the generating functions coincide ($G_D(z) \approx G_C(z)$), and the entire macroscopic density remains unaltered. It is precisely this spectral conservation that confers stability to the algorithm: by preserving the covariance structure in the space of time and frequencies, it is guaranteed that the denoiser can operate on the true patterns of the signal without being distracted by microscopic reshuffling.

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